Chapter 1

Helium Step Heat Experiment on Goethite in Hydrothermal Conditions Indicates High Retentivity at Surface Temperature and is Consistent with Multiple Diffusion Domain Systematics

1.1 Abstract

U-Th/He geochronology of goethite (FeOOH) is a powerful method to quantitatively date weathering and erosion processes on Earth's surface; however, precise interpretation of goethite U-Th/He dates is challenging because uncertainty persists regarding the mineralogical and environmental conditions under which radiogenic helium is retained and how it may be lost over time. Standard vacuum step heating experiments have been successfully used on a variety of mineral phases to characterize helium migration, but the results of such experiments on goethite predominantly record dehydroxylation-induced helium loss, raising questions about their relevance to helium retention and loss in nature (Farley, H. B. Monteiro, et al., 2024). Observations that thermal decomposition of goethite is greatly suppressed in the presence of high water pressure led us to develop a new method to conduct helium step heat experiments under water-saturated conditions up to 300°C within standard experimental setups. Using this method, we report results from the first helium step heat experiment of proton-irradiated goethite under hydrothermal conditions. Irradiated goethite sample Win-06-01b of the Flinders Ranges, South Australia and distilled water were encapsulated in a pure silica glass ampoule and were subjected to multiple prograde and retrograde thermal cycles over a temperature range of 170°C to 250°C under water vapor saturated conditions with step durations ranging from 2 hours to over a week. The results show apparent helium diffusivities approximately seven log units lower than those observed in vacuum, converging to a reproducible linear array on an Arrhenius plot with an activation energy of 157.1± 5.3kJ/mol and a closure temperature of 98.1± 8.6°C at a cooling rate of 10°C/Ma. These results are consistent with a volume-diffusion helium loss mechanism and show high helium retentivity for natural goethite that extrapolates to near-quantitative helium retention at surface conditions over billion-

year timescales. A four-domain diffusion model fitted to the ^{3}He release places 97% of the ^{3}He in a single domain and attributes the natural loss of radiogenic helium from that domain to thermally activated diffusion. Steady cooling from crystallization at ∼76–80 °C in the Jurassic fits the ^{4}He/^{3}He spectrum best, favoring a hydrothermal origin, while burial of surface-formed goethite to ∼60 °C in the Late Cretaceous remains a viable alternative. Reanalysis of published vacuum step heats of the same sample shows that they record first-order conversion to hematite, whereas the hydrothermal release records diffusion. Thermodynamic and nucleation-kinetic modeling calibrated on vacuum step heat experiments on the same sample explains this difference and shows that conducting step heats with this method is expected to slow dehydroxylation of Win-06-01b by many orders of magnitude, effectively suppressing dehydroxylation on experimental timescales. This analysis applies only to Win-06-01b, and can only be extended to other goethite samples for which kinetic data or vacuum step heat data are available due to the diversity and complexity of goethite morphologies and grain sizes. The method presented in this work can be systematically applied to experimentally investigate effects of crystallite size, chemical composition, and radiation damage to refine these results and to assess the applicability of existing vacuum step heat data to helium retention and loss in goethite in nature.

1.2 Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024). Goethite is a powerful chronometer for dating surface processes such as weathering and ferruginization directly: laterite and duricrust formation and recrystallization (H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Allard et al., 2018; Ansart et al., 2022), and the ferruginization of organic matter and paleosols (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann et al., 2017).

Interpreting a goethite U-Th/He date requires knowing the conditions under which the mineral retains radiogenic helium. As with any radiochronometer, meaningful geological interpretation of a measured age depends on knowing when the system is open to loss or gain of parent or daughter nuclides. For goethite, step heats of proton-irradiated, crystalline samples are used to reveal the degree of partial helium loss by the 4He/3He method (Shuster, P. M. Vasconcelos, et al., 2005), and suggest that helium loss fractions in nature are always non-zero but modest, ranging from 2 to 30% (Shuster, P. M. Vasconcelos, et al., 2005; Heim et al., 2006; Hofmann et al., 2017; Deng et al., 2017). Although one study attempts to explain the variable loss principally as a result of variable radiation damage (Bassal et al., 2022), the effects of temperature, crystallite size, and chemical composition have not been determined experimentally and remain unresolved, because goethite decomposes to hematite (Fe_2O_3) in the step heat experiments used to characterize it (Farley, H. B. Monteiro, et al., 2024).

Laboratory experiments of helium migration under controlled conditions have proven extremely useful in understanding the mechanisms and rates of helium migration in geological materials and have produced sensible results that are predominantly explainable by thermally activated volume diffusion (Reiners and Farley, 1999; Farley, 2000; Farley, 2007; Cherniak and Watson, 2011; Farley, 2018), albeit variably complicated by the effects of radiation damage and annealing (Shuster, Flowers, et al., 2006; Guenthner et al., 2013). This approach has also been leveraged for goethite; however, some complications arise when applying this approach to goethite. Above surface temperatures, goethite decomposes to hematite and water, often along crystallite elongation directions (Goss, 1987). Farley, H. B. Monteiro, et al. (2024) measured evolved water by capacitance manometry during vacuum step heats and found helium and water release to be tightly correlated even

below 200°C, indicating that decomposition, rather than helium migration through an unchanging crystal, governs the release. Vacuum step-heat results for goethite therefore need not describe helium behavior in nature.

Still, existing helium step heat measurements of goethite in vacuum likely remain informative to an extent, as goethite dehydration kinetics are highly influenced by some of the same factors that could be responsible for variable helium retention in natural goethite, like variable grain size (Navrotsky et al., 2008; Guo and Barnard, 2011; Goss, 1987), crystallinity (Schwertmann et al., 1985) or substitution of aluminum or other cations within the goethite structure (Majzlan, Grevel, et al., 2003; Bassal et al., 2022). For example, Shuster, P. M. Vasconcelos, et al. (2005) initially conducted vacuum step-heating experiments on two vitreous, polycrystalline goethite samples, in which uniformly distributed 3He was introduced by proton irradiation for comparison with radiogenic 4He that was partially lost in nature. The results of Shuster, P. M. Vasconcelos, et al. (2005) could be well-fit with a model of two distinct diffusion domains or migration mechanisms with significant differences in helium retention under surface conditions, e.g. crystallite interiors and intergranular space. Although these results may have been impacted by dehydration in vacuum, certain aspects may remain applicable to helium loss in goethite in nature if helium loss in nature and the laboratory advance through the samples with similar geometries. One important point is that if laboratory experiments degas domains in order of their helium retentivity in nature, even if the rates of loss are affected by goethite dehydration, then a $^4He/^3He$ experiment in vacuum can still be used to measure helium loss (Shuster, P. M. Vasconcelos, et al., 2005; P. M. Vasconcelos et al., 2013).

An additional complexity is that natural goethite almost always occurs as aggregates of minute crystallites with cross-sections smaller than 1 µm in nature. Neglecting to account for multiple diffusion domains can lead to inaccurate determinations of activation energy (E_a) and frequency factor (D_o/a^2) for helium diffusion (Farley, 2018). Detailed step-heating with temperature cycling, as has been done for hematite (Farley and Flowers, 2012; Farley, 2018), is needed to characterize domain size and activation energy for polycrystalline samples.

Molecular simulations indicate that helium diffusion in perfectly crystalline goethite should be highly anisotropic, essentially reducing to 1-dimensional diffusion along the b-axis (typically the long axis of the polycrystalline aggregate) that would amount to near-instantaneous helium escape from goethite (Bassal et al., 2022), which is clearly contradicted by observed (U–Th)/He dates as old as ~275 Ma (Waltenberg, 2012) and dates that are concordant with $^{40}Ar/^{39}Ar$ ages of other phases (P. M. Vasconcelos et al., 2013). However, the models presented by Bassal et al. (2022) suggest that crystallographic defects caused by radiation damage could block these channels, thereby reducing He loss. While these simulations are informative, they alone are insufficient for drawing conclusions about the mineralogical and environmental conditions needed for accurately dating the crystallization of goethite with the (U–Th)/He method and require experimental verification.

Goethite dehydration is suppressed at high water pressure (Langmuir, 1971; Navrotsky et al., 2008), so a step heat conducted under hydrothermal conditions may interrogate helium loss rather than decomposition kinetics. Until now, no studies have reported helium step-heating results for goethite under hydrothermal (high pH_2O) conditions, which could help stabilize goethite or slow its decomposition to illuminate the processes relevant to helium loss in natural settings. If experimental conditions can be controlled to slow the decomposition of goethite such that the rate of natural helium loss in goethite greatly exceeds the rate of goethite decomposition into hematite and water, it would open up an experimental approach to systematically assess how factors like crystallite size, temperature, radiation damage, and chemical composition could affect helium mobility and retentivity in nature.

Early trials of the method were encouraging (§1.3.3). SEM imagery taken of thermally treated

natural goethites in vacuum and open air showed the development of contraction cracks on the outside and throughout the interior of goethite grains heat treated in vacuum and in open air (Figs. 1.1, 1.3), while goethite heat-treated when encapsulated with water or gypsum did not form cracks (Fig. 1.2). Goethite heat treated in open air turned red, whereas goethite heat treated in vacuum remained black.

We test this premise with the first helium step heat experiment on goethite under water-saturated conditions, and compare it with published vacuum step heats of the same sample (§1.5.1–1.5.2). A thermodynamic and kinetic analysis of goethite dehydroxylation then explains why water has this effect (§1.5.3–1.5.4).

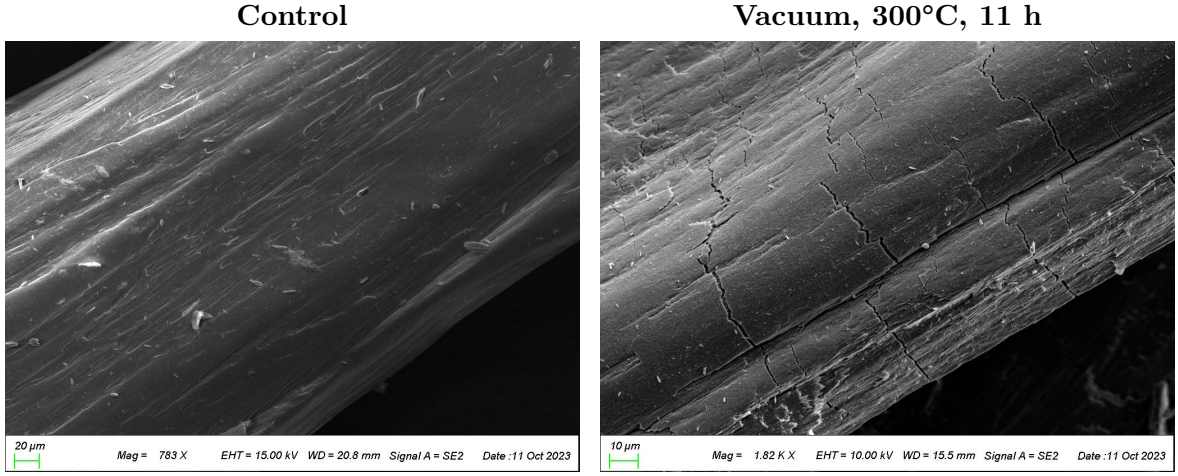


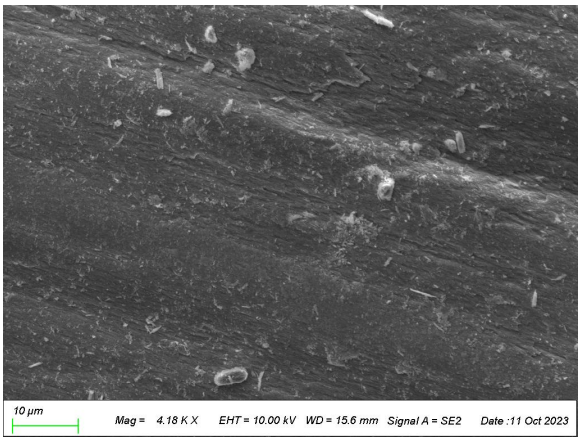
Figure 1.1: Secondary electron images of Bahia goethite aggregate before heat treatment (control) and after heat treatment at 300°C for 11 hours in vacuum. Cracks developed in the vacuum-heated sample both parallel and perpendicular to the long axis of the polycrystalline aggregate, representing volume contraction due to dehydration/decomposition of goethite into hematite. The buffered treatments at the same temperature are shown in Fig. 1.2.

1.3 Experimental Methods

This study builds upon previous helium step heat studies of the dense colloform goethite sample Win-06-01b (occasionally referred to as Win0601b) collected within float at the Flinders Ranges in South Australia (Waltenberg, 2012; H. Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages: 180-135 Ma (Waltenberg, 2012). To produce a homogeneous distribution of 3He, Win-06-01b was irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster and Farley, 2005) and subsequently was subjected to helium step heat measurements in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024). In their step heat experiment of Win-06-01b, Farley, H. B. Monteiro, et al. (2024) additionally measured evolved water using capacitance manometry to show that helium and water release were tightly correlated for goethite step heats in vacuum. Although they used subtly different heating schedules, the results of Farley, H. B. Monteiro, et al. (2024) clearly agree well with H. Monteiro (2017).

In this study, an aliquot of the same irradiated Win-06-01b goethite used by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) was encapsulated with MilliQ water within a silica glass tube. Silica glass was chosen over other laboratory glasses such as borosilicate because it is the

Gypsum-buffered, 300°C, 8 h



Water-buffered, 300°C, 4 h

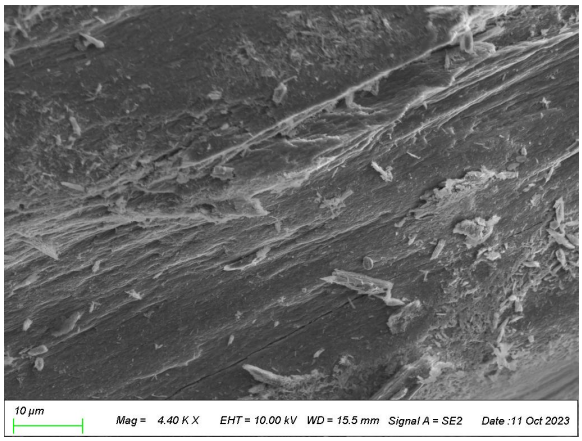


Figure 1.2: Secondary electron images of Bahia goethite aggregate heated at 300°C in a sealed glass tube with gypsum (8 hours) and with water (4 hours). Despite treatment at high heat, no cracks indicative of dehydration were visible on either goethite surface. In the gypsum experiment, the gypsum dehydrated first and produced enough water pressure to stop or slow dehydration of the goethite and prevent the development of microscopic cracks; at higher temperatures goethite and gypsum reacted, forming a surface layer of other minerals. For the water-buffered sample, no cracks were observed when investigating the goethite at scales similar to Fig. 1.1, and most of the aggregate's surface resembled the gypsum-buffered panel; the area pictured had the roughest texture on the grain. A very small crack parallel to the long axis may indicate some amount of dehydration had occurred, or it could have been inherited from before hydrothermal treatment, since even the untreated control grain (Fig. 1.1) had some rough surfaces and rare long-axis-parallel cracks.

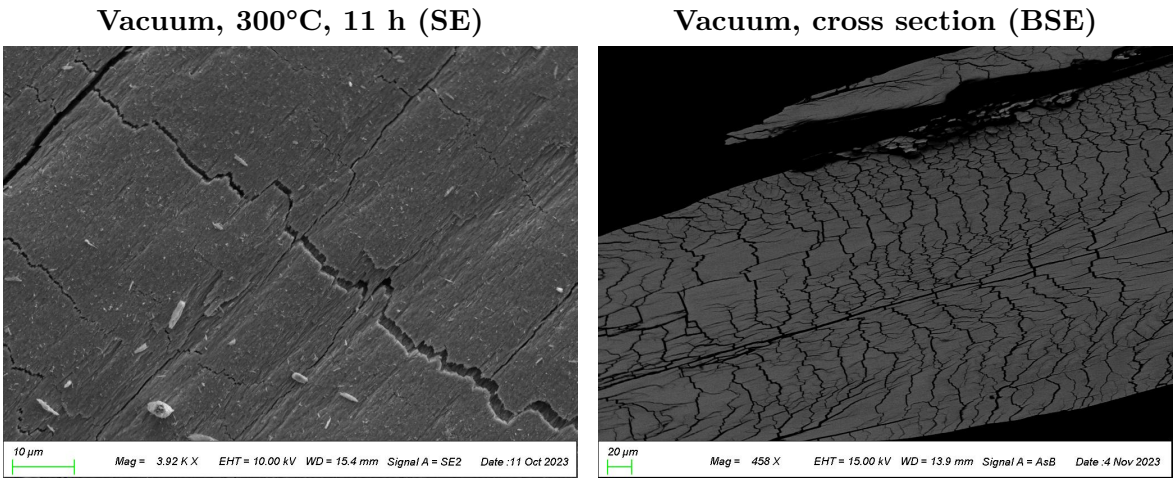


Figure 1.3: Vacuum-heated Bahia goethite at higher magnification and in cross section. Left: secondary electron image of goethite heated at 300°C for 11 hours in vacuum, at the same scale as the buffered samples of Fig. 1.2, showing rough cracks at various scales both parallel and orthogonal to the long axis of the goethite aggregate. These are the typical decomposition features that would be expected in Fig. 1.2 if significant decomposition had occurred there. Right: backscatter electron image of a polished cross section of epoxy-mounted, vacuum-heated Bahia goethite, from the same subsample as the vacuum-heated grain of Fig. 1.1, exposing the grain's interior. This image clearly shows that the cracks visible on the exterior of dehydrated goethite are pervasive throughout the interior as well.

most permeable glass type to helium (Norton, 1953; Altemose, 1961). To prepare the silica glass tube, 10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thinner pipettes using a natural gas and oxygen glass-welding torch and the pipette ends were fused closed. After the glass had cooled, 1.18 mg of irradiated Win-06-01b goethite blades less than 2 mm in length were carefully transferred to the bottom of the tube in the presence of an ionizing fan. Less than a microliter of distilled water was transferred to the sealed pipette using a thin glass capillary, submerging the goethite, after which the tube was gently tapped to liberate the bubbles and ensure that the water and goethite were efficiently packed at the bottom of the pipette. To seal the pipette, the contents of the ampoule were flash-frozen in liquid nitrogen and transferred to a saltwater-ice bath at a temperature of -10°C. While the pipette end of the tube remained frozen in the saltwater-ice bath, the top was interfaced with a vacuum line and the tube was pumped to a pressure of $<$0.1 torr. The advantage of this approach was that water would remain solid within the tube, only slowly sublimating to vacuum while atmospheric gases would be readily pumped away. While it was desirable to remove atmospheric gases to reduce experimental complexity by removing potentially reactive gases like oxygen, the primary motivation for pumping the tube to rough vacuum was to aid the sealing process. When the tube contents were under vacuum, fused glass would deform inwards and self-seal, greatly improving the ease and quality of the seal while requiring far less heat input from the torch. After an acceptably low pressure was attained, the tube was closed to the vacuum line and the saltwater-ice bath was quickly replaced with a liquid nitrogen bath, ensuring that no ice was melted. The liquid nitrogen bath was preferable during sealing as it more effectively protected the goethite and ice from the high temperatures of the

glass-welding torch. A fine-flamed welding torch was used to seal the top of the ampoule, which dropped into the liquid nitrogen bath and was retrieved by passing the liquid nitrogen through a sieve. Fig. 1.4 shows the sealed ampoule after the completion of the step heats with all of the original water remaining, a testament to the quality of the ampoule's seal, which had sustained a massive pressure gradient of over 11 orders of magnitude across the ampoule walls, with an inner vapor pressure of about 3×10^4 torr (~40 bars: Haar et al. (1984)) against the ~10^{-7} torr ultra-high vacuum of the lamp cell while the ampoule was held at 250°C for over a week.



Figure 1.4: A view of the entire experimental ampoule after the completion of the step heat experiment before it was broken for the final step in the resistance furnace. At room temperature, the tube is approximately half-filled with gas, one-third filled with liquid water, and one-sixth filled with <2 mm long shards of irradiated Win-06-01b goethite. A gel-like substance was observed lining the inner walls of the tube, which is likely to be amorphous silica dissolved from the tube and redeposited by the thermally-cycled, high pressure liquid water phase.

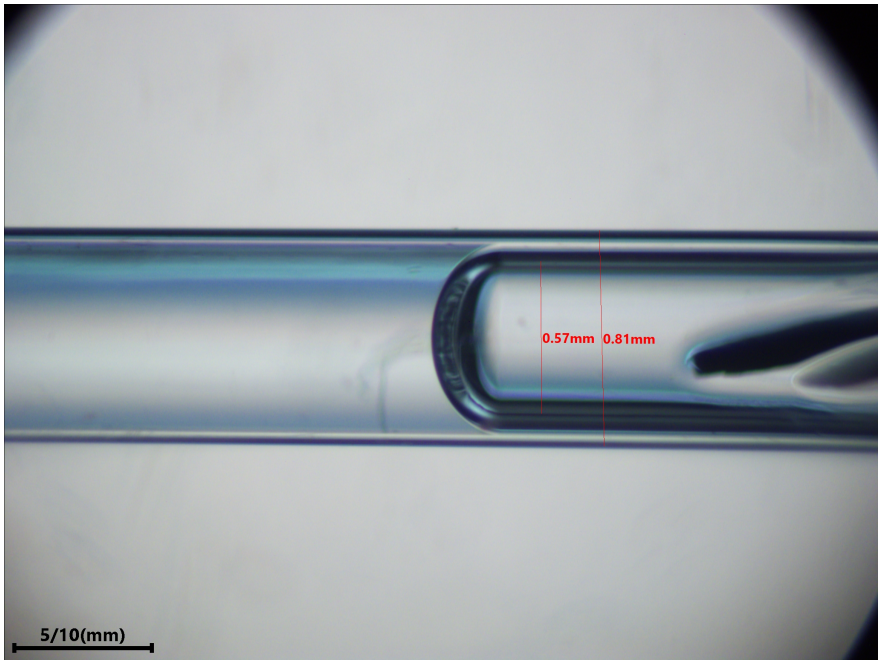


Figure 1.5: Measurements of the inner and outer diameter of the ampoule used in the hydrothermal step heats of Win-06-01b. The inner walls of the ampoule are only clearly visible within air bubbles.

The sealed ampoule, now referred to as HT-Win-06-01b, was placed in a copper foil packet and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley, Reiners, et al. (1999). Before the experiment began, the ampoule was loaded in the lamp cell, pumped to high vacuum, and was held at 70°C for 4 days to degas atmospheric helium entrained within the ampoule.

HT-Win-06-01b was subjected to multiple prograde-retrograde heating cycles to investigate possible MDD behavior. Step durations varied from 2 hours to about a week; the total experiment was conducted over about 40 days and included experimental steps ranging from 170-250°C. We elected not to heat the sample hotter because high temperatures generated very high vapor pressures. In early trials, we found that similar ampoules most often withstood the ~86 bar vapor pressures associated with liquid water saturation at 300°C (Haar et al., 1984), but they would explode when further heated to temperatures beyond 315°C.

Experimental temperature records show ~1°C amplitude oscillations around the target temperature. While we cannot exclude the possibility that sustained temperature gradients were present throughout the sample due to the use of the comparatively large ~1 mm thick ampoule, we are confident that the conclusions of this work are robust. The ampoule was in direct contact with the thermocouple junction and was in front of the thermocouple junction with respect to the projector lamp heating source, suggesting that if there were a temperature discrepancy, the sample would have been hotter than the recorded temperature, which would not affect the conclusions of this work. Additionally, during early trials of this method for helium step heats, silica-glass encapsulated Durango apatite had been subjected to step heats at Caltech and gave corrected 4He yields indistinguishable from those of unencapsulated Durango apatite, suggesting that the temperature recorded by the thermocouple for our experimental setup is sufficiently accurate.

Helium isotope measurements of evolved gases from hydrothermal step heats of glass-encapsulated irradiated HT-Win-06-01b goethite were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from January 31st, 2024 to March 13th, 2024. Helium was purified by exposing evolved gases to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray, which was never significant during the experiment. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility.

A cold blank was measured at the beginning of the experiment, showing a negligible 3He blank and a 4He blank signal corresponding to about 1.25 pccSTP, which was similar to 4He values attained during low temperature permeation step measurements. Permeation steps were thus assumed to measure shifting 4He blanks during the experiment, and the entire blank correction was added to all 4He uncertainties.

For the final step, the HT-Win-06-01b ampoule was broken and dried overnight in a 50°C oven, then completely degassed by heating to ~1100 °C in a resistance furnace. The final step measurement was conducted on a Thermo Fisher Helix SFT mass spectrometer. A hot furnace blank subjected to the same volume splits as the final HT-Win-06-01b measurement gave a 3He blank of 6.8 fccSTP and a 4He blank of 0.31 nccSTP. This blank correction was less than 1% of the measured

4He and negligible for 3He. A re-extract of HT-Win-06-01b immediately following the measurement yielded above-blank 3He and 4He signals, which were added to the total measurement, although both constituted less than 1% of the final step yield. Sensitivity was calculated by measuring a 3He-enriched air standard with known 3He and 4He contents immediately before the measurement.

Helium released by the sample reaches the mass spectrometer only after permeating the ampoule wall, so each measured step yield lags the true release. Step yields were corrected for this permeation lag with the bounding method of Chapter **??** (§**??**), which gives lower-bound, central and upper-bound values for every step (Tables 1.1 and 1.2). The correction and the ampoule permeabilities measured during the experiment are described in the Supplementary Information (§**??**), together with the diffusivity inversions, the isotope-ratio uncertainties and the kinetic analysis of the published vacuum step heats (§**??**).

1.3.1 Kinetic and Diffusive Reductions of Step-Heat Data

We reanalyzed the published vacuum step heats of irradiated Win-06-01b by Farley, H. B. Monteiro, et al. (2024) (hereafter *KAF*) and H. Monteiro (2017) (hereafter *HDM*), together with the water-saturated step heat of this study, by reducing every step both as a first-order conversion rate constant and as an apparent diffusivity. For the kinetic reduction, ^{3}He is assumed to be immediately released from the goethite lattice upon conversion to hematite, except for a fraction ϕ that is inherited by the hematite product. The fraction of helium released in each step is then $F_i = (1-\phi)\,\alpha_i$, where α_i is the fraction of goethite converted in step i.

We describe the kinetics of goethite dehydroxylation with a first-order rate law,

$$\frac{d\alpha}{dt} = k(T)\,(1-\alpha), \qquad k(T) = A\exp\!\left(-\frac{E_a}{RT}\right), \tag{1.1}$$

For a schedule of isothermal steps of duration Δt_j at T_j, Eq. (1.1) integrates to

$$\alpha_i = 1 - \exp\left[-A\sum_{j\le i},\Delta t_j \exp\!\left(-\frac{E_a}{RT_j}\right)\right], \tag{1.2}$$

Inverting Eq. (1.2) step by step gives an apparent rate constant for every step that is independent of any fitted parameter except ϕ:

$$k_i = -\frac{1}{\Delta t_i}\ln\frac{1-\alpha_i}{1-\alpha_{i-1}}, \qquad \alpha_i = \frac{F_i}{1-\phi}. \tag{1.3}$$

For comparison, the diffusion parameter of each step is obtained with the same Fechtig and Kalbitzer (1966) inversion for a sphere that is applied to the hydrothermal experiment (§**??**).

Additionally, in both experiments release stalls at $F \approx 0.9$. The fitted $\phi = 0.101$ is the same for KAF and HDM. In HDM, the remaining ∼10% is released slowly between 340 and 500 °C and on fusion. We interpret this fraction as ^{3}He inherited by the hematite product and subsequently released by the much slower diffusion out of hematite. The conversion and the plateau correspond to the "Stage 2" and "Stage 3" release of Farley, H. B. Monteiro, et al. (2024), whose transition near $F \approx 0.9$ is independent of heating schedule (and tracked by a distinct shift in ^{4}He/^{3}He: Fig. 1.15). The kinetic inversions (Eq. (1.3)) for these latest steps are highly sensitive to the choice of ϕ and are excluded from the fits of the kinetic parameters. The KAF steps run with the trap isolated are also excluded from the Arrhenius regression.

1.3.2 Multiple Diffusion Domain and Thermal History Modeling

Together, the ^{3}He release and the bulk-normalized ^{4}He/^{3}He spectrum constrain the thermal history, if the activation energy for diffusion is known (Farley and Flowers, 2012). These data from HT-Win-06-01b are interpreted in two stages. First, the diffusion domain structure of the sample is inverted from the ^{3}He release alone. Second, natural thermal histories are fitted to the ^{4}He/^{3}He spectrum with that domain structure held fixed, so that every thermal history must work within the bounds set by the ^{3}He data. Derivations and numerical details are given in the SI (§**??**).

Diffusion domain structure. The sample is represented by N non-interacting spherical domains that share the activation energy and frequency factor of the Stage 1 regression (§1.4.1). Domain j holds a fraction f_j of the uniformly distributed, proton-induced ^{3}He ($\sum_j f_j = 1$) and has radius r_j relative to the reference radius r_0 implied by the Stage 1 regression, so that

$$\left(\frac{D}{a^2}\right)_j = \left(\frac{r_j}{r_0}\right)^{-2} \frac{D_0}{a_0^2} \exp\!\left(-\frac{E_a}{RT}\right). \tag{1.4}$$

A model with N domains has $2N-1$ free parameters. During the heating schedule, the reference domain accumulates the non-dimensional time $\tau_{0,i} = \sum_{m\le i}(D_0/a_0^2)\exp(-E_a/RT_m)\,\Delta t_m$ by the end of step i, and domain j reaches $\tau_{0,i}(r_j/r_0)^{-2}$. With $F(\tau)$ the cumulative fraction released by an initially uniform sphere (Crank, 1979), the model cumulative and fractional ^{3}He yields are

$$\Sigma F_i = \sum_{j=1}^{N} f_j\, F\!\left(\tau_{0,i}\,(r_j/r_0)^{-2}\right), \qquad \Delta F_i = \Sigma F_i - \Sigma F_{i-1}. \tag{1.5}$$

Following Farley and Flowers (2012), the fractional and cumulative yields are weighted equally, and $\{f_j, r_j\}$ minimize

$$S = \sum_i \left(\Delta F_i^{\mathrm{mod}} - \Delta F_i^{\mathrm{obs}}\right)^2 + \sum_i \left(\Sigma F_i^{\mathrm{mod}} - \Sigma F_i^{\mathrm{obs}}\right)^2, \tag{1.6}$$

summed over the 28 steps with more than 0.1 pccSTP of ^{3}He. The gas released in the six low-signal steps still counts toward the cumulative yields of later steps, and their durations toward the heating schedule. Domains are added one at a time, each search starting from the $(N-1)$-domain solution, until an added domain no longer changes the fit or the domain structure. The Arrhenius and delta plots are not fitted: they follow from applying the Fechtig and Kalbitzer (1966) inversion to the model cumulative release, and their misfit is reported as a reduced χ^2 over the same 28 steps.

Thermal histories. Any thermal history predicts a ^{4}He/^{3}He spectrum that is obtained following the procedure of Shuster, Farley, et al. (2004). A summary of this procedure is given in the SI (§**??**). Thus, we evaluate how well each thermal history reproduces the measured ^{4}He/^{3}He spectrum. We compare three types of thermal histories, with two free parameters for steady cooling and the isothermal hold and three for burial and exhumation.

- **Steady cooling:** the temperature falls linearly from T_0 at t_{form} to 25 °C today. The goethite forms at an elevated temperature, for example from warm fluids at depth, and cools monotonically. Free parameters: t_{form} and T_0.
- **Burial and exhumation:** the path passes through (t_{form}, T_s), $(t_{\mathrm{peak}}, T_{\max})$, (50 Ma, T_s) and $(0, T_s)$ with $T_s = 20$ °C. For this model, we assume that the goethite forms at the surface, is buried and heated to a peak temperature, and is exhumed by the Paleogene, roughly following

the thermal history (Gibson and Stüwe, 2000; Mitchell et al., 1998) interpreted for the region from apatite fission track thermochronometry. We fit t_{peak} and $T_{\max}$ at each of three fixed crystallization ages of 180, 195 and 210 Ma; because the crystallization age is also varied across the three fits, burial has three free parameters.

- **Isothermal hold:** $T = T_{\text{iso}}$ from t_{form} to the present. Free parameters: t_{form} and T_{iso}.

Each path is scored against

$$\chi^2 = \sum_i \left(\frac{R_i^{\text{mod}} - R_i^{\text{M}}}{\sigma_i} \right)^2, \qquad \sigma_i = \max\left[\tfrac{1}{2} \left(R_i^{\text{UB}} - R_i^{\text{LB}} \right), 0.02 \right], \tag{1.7}$$

over the 24 steps that have lag-correction bounds and more than 0.1 pccSTP of ^{3}He. Where M, LB and UB are the central, lower-bound and upper-bound bulk-normalized ratios of the bounding method (Table 1.2). This analysis fits 24 independent data points; subtracting the fit parameters gives $\nu = 22$ degrees of freedom for steady cooling and the isothermal hold and $\nu = 21$ for burial and exhumation, which are used to evaluate the reduced χ^2. χ^2 is evaluated for these free parameters along regularly spaced grids to produce a likelihood heatmap (Fig. 1.17). The measured He age (130–180 Ma; Waltenberg 2012) is treated as a hard constraint. The best fit of each family is the lowest-χ^2 path with a predicted age in that range, and parameter ranges are those of the age-consistent paths within the same $\Delta\chi^2$ of that best fit.

1.3.3 SEM Study of Thermally Treated Goethite

In early trials of the "glass tube method" for helium step heat experiments with controlled microenvironments presented here, sub-samples of goethite morphologically similar to Win-06-01b from Igarapé Bahia (Shuster, P. M. Vasconcelos, et al., 2005; H. Monteiro, 2017) were heated in vacuum using the projector lamp heating apparatus of Farley, Reiners, et al. (1999) and in sealed glass ampoules with gypsum or water additionally added to provide moisture to slow/prevent dehydration of goethite. All samples were held at 300°C for 4-11 hours, and the glass ampoules were broken and the goethites were allowed to air dry before all samples were placed on carbon tape and carbon coated. These samples were imaged at Caltech in secondary electron mode by the ZEISS 1550VP Field Emission SEM. Synthetic goethite pigment, a commercial yellow iron-oxide pigment of acicular crystals ∼1 µm long, was treated in the same way but only with water, without vacuum or open-air runs: at least five larger silica glass ampoules, each loaded with pigment and an excess of water and fused shut at ambient pressure, were held at 200–250 °C for about a day in an oven whose voltage was adjusted by hand. These runs were a rapid, qualitative test; details of individual runs (temperature histories, durations, water-to-solid ratios) were not recorded, and reaction was assessed by color (yellow goethite, red hematite) before imaging.

1.4 Results

1.4.1 Water-Saturated Step Heat of HT-Win-06-01b

The complete corrected step-heat dataset underlying the diffusivities, delta plots, and Arrhenius regressions discussed below is reported in Tables 1.1 and 1.2, including the lower-bound, central, and upper-bound values from the bounding method's permeation-lag correction for ^{3}He and ^{4}He respectively.

Table 1.1: ^{3}He step-heat dataset for HT-Win-06-01b under the bounding method: measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), the lower- and upper-bound true step production (ω_{LB}, ω_{UB}), and the corresponding lower-bound, central, and upper-bound apparent diffusivities. "–" denotes early steps with $\lambda t < 4$, for which bounds could not be determined (see §**??**). Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots (Fig. 1.15 and others derived from this dataset).

Temp	Time	^{3}He meas.	Cum. ^{3}He	$n_{t,\mathrm{UB}}$	ω (pccSTP)		$\ln(D/a^2)$		
(°C)	(min)	(pccSTP)	(pccSTP)	(pccSTP)	LB	UB	LB	m	UB
170	300	0.9630	0.0000	1.1660	–	–	–	–	–
180	463	0.5442	0.9630	0.2413	–	–	–	–	–
190	550	0.5292	1.6241	0.1538	–	–	–	–	–
200	500	0.5190	2.2112	0.1430	0.3651	0.6619	-22.16	-21.77	-21.50
210	475	0.5770	2.7646	0.1440	0.4341	0.7210	-21.71	-21.41	-21.16
220	450	0.6333	3.3799	0.1447	0.4894	0.7781	-21.34	-21.07	-20.84
230	425	0.7899	4.0551	0.1685	0.6452	0.9584	-20.82	-20.60	-20.39
225	908	0.7513	4.8885	0.0716	0.5827	0.8229	-21.52	-21.25	-21.15
215	450	0.1415	5.6842	0.0349	0.0699	0.1764	-22.84	-22.12	-21.90
205	520	0.0763	5.8529	0.0184*	0.0413*	0.0947*	-23.48*	-22.86*	-22.65*
195	1106	0.0670	5.9491	0.0080*	0.0487*	0.0750*	-24.05*	-23.73*	-23.62*
185	740	0.0227	6.0246	0.0049*	0.0147*	0.0276*	-24.84*	-24.40*	-24.21*
175	920	0.0180	6.0610	0.0036*	0.0131*	0.0216*	-25.17*	-24.85*	-24.66*
200	260	0.0224	6.0814	0.0163*	0.0189*	0.0388*	-23.53*	-23.36*	-22.81*
210	520	0.0812	6.1081	0.0181*	0.0649*	0.0993*	-22.98*	-22.76*	-22.55*
220	1167	0.5766	6.1914	0.0447	0.5585	0.6213	-21.59	-21.55	-21.48
230	370	0.3577	6.7796	0.0903	0.3130	0.4480	-20.95	-20.81	-20.58
240	320	0.6767	7.1563	0.1776	0.5864	0.8543	-20.06	-19.91	-19.67
245	250	0.5549	7.8582	0.1888	0.3773	0.7437	-20.18	-19.78	-19.47
250	210	0.6934	8.4743	0.2786	0.5047	0.9721	-19.63	-19.30	-18.94
245	270	0.4659	9.2086	0.1413	0.1872	0.6071	-20.80	-19.87	-19.60
240	320	0.2857	9.7206	0.0749	0.1445	0.3606	-21.18	-20.49	-20.25
235	370	0.2199	10.0357	0.0517	0.1450	0.2716	-21.28	-20.86	-20.65
230	420	0.1400	10.2736	0.0303	0.0883	0.1703	-21.89	-21.42	-21.22

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Temp	Time	^{3}He meas.	Cum. ^{3}He	$n_{t,\mathrm{UB}}$	ω (pccSTP)		$\ln(D/a^2)$		
(°C)	(min)	(pccSTP)	(pccSTP)	(pccSTP)	LB	UB	LB	m	UB
225	652	0.1494	10.4279	0.0206	0.1191	0.1700	-22.01	-21.78	-21.65
230	500	0.1550	10.5908	0.0271	0.1344	0.1820	-21.61	-21.46	-21.30
235	450	0.1951	10.7582	0.0361	0.1681	0.2313	-21.26	-21.11	-20.94
240	400	0.2425	10.9638	0.0483	0.2063	0.2908	-20.91	-20.75	-20.57
245	350	0.3150	11.2328	0.0690	0.2667	0.3840	-20.50	-20.33	-20.12
250	300	0.3716	11.5689	0.0939	0.3026	0.4656	-20.18	-19.97	-19.74
250	300	0.3640	11.9627	0.0899	0.2700	0.4539	-20.26	-19.96	-19.73
250	300	0.3452	12.3490	0.0852	0.2553	0.4304	-20.29	-19.98	-19.75
250	10250	8.4829	12.7167	0.0494	8.3977	8.5324	-19.99	-19.98	-19.97
240	785	0.2261	21.2187	0.0209	0.1767	0.2470	-21.03	-20.78	-20.69
Total			146.1483						

Table 1.2: ^{4}He step-heat dataset and absolute ^{4}He/^{3}He ratios for HT-Win-06-01b under the bounding method, structured as in Table 1.1. "–" denotes steps for which bounds could not be determined. The Total row reports the bulk (total-yield) ^{4}He/^{3}He ratio, i.e. the normalizing value used to bulk-normalize these ratios for the plot in Fig. 1.15. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots, as in Table 1.1.

Temp	Time	^{4}He meas.	Cum. ^{4}He	$n_{t,\mathrm{UB}}$	ω (nccSTP)		$\ln(D/a^2)$			^{4}He/^{3}He		
(°C)	(min)	(nccSTP)	(nccSTP)	(nccSTP)	LB	UB	LB	m	UB	LB	m	UB
170	300	0.0192	0.0000	0.0280	–	–	–	–	–	–	–	–
180	463	0.0305	0.0192	0.0152	–	–	–	–	–	–	–	–
190	550	0.0558	0.0660	0.0179	–	–	–	–	–	–	–	–
200	500	0.1024	0.1375	0.0313	0.0845	0.1337	-26.21	-25.96	-25.62	–	197	–
210	475	0.1460	0.2500	0.0404	0.1147	0.1864	-25.31	-25.02	-24.72	248	253	269
220	450	0.2091	0.4055	0.0530	0.1686	0.2620	-24.41	-24.15	-23.88	326	330	349
230	425	0.3243	0.6319	0.0760	0.2714	0.4004	-23.43	-23.21	-22.96	404	411	432
225	908	0.3483	0.9799	0.0363	0.2722	0.3845	-23.81	-23.53	-23.41	443	464	474
215	450	0.0716	1.3487	0.0196	0.0353	0.0912	-24.95	-24.23	-23.98	429	506	537
205	520	0.0385	1.4359	0.0103*	0.0189*	0.0487*	-25.66*	-24.94*	-24.70*	392*	504*	535*
195	1106	0.0336	1.4912	0.0043*	0.0233*	0.0379*	-26.16*	-25.80*	-25.68*	447*	501*	515*
185	740	0.0079	1.5467	0.0019*	0.0037*	0.0098*	-27.59*	-26.81*	-26.59*	225*	350*	369*
175	920	0.0152	1.6020	0.0033*	0.0133*	0.0185*	-26.48*	-26.34*	-26.14*	843*	843*	982*
200	260	0.0107	1.6351	0.0091*	0.0073*	0.0198*	-25.79*	-25.41*	-24.79*	376*	476*	555*
210	520	0.2166	1.6547	0.0535*	0.2075*	0.2701*	-23.07*	-23.02*	-22.79*	2667*	2667*	3104*
220	1167	0.3187	1.8714	0.0269	0.2652	0.3456	-23.50	-23.30	-23.21	472	553	563
230	370	0.2113	2.6766	0.0590	0.1844	0.2703	-22.39	-22.25	-21.99	575	591	628
240	320	0.4286	3.1616	0.1243	0.3697	0.5529	-21.36	-21.20	-20.93	612	633	675
245	250	0.3988	3.6430	0.1521	0.2746	0.5510	-21.29	-20.90	-20.56	666	719	778
250	210	0.4440	4.2359	0.2020	0.2918	0.6460	-20.91	-20.47	-20.07	542	640	703
245	270	0.3123	4.7263	0.1053	0.1103	0.4176	-22.04	-20.98	-20.68	469	670	721
240	320	0.1927	5.0846	0.0559	0.0874	0.2485	-22.38	-21.57	-21.32	506	674	718
235	370	0.1517	5.3031	0.0393	0.0958	0.1910	-22.39	-21.92	-21.69	600	690	730
230	420	0.0991	5.4714	0.0236	0.0598	0.1227	-22.96	-22.45	-22.23	608	708	746
225	652	0.0995	5.5835	0.0151	0.0760	0.1146	-23.14	-22.86	-22.72	608	666	689

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Temp	Time	^{4}He meas.	Cum. ^{4}He	$n_{t,\mathrm{UB}}$	ω (nccSTP)		$\ln(D/a^2)$			^{4}He/^{3}He		
(°C)	(min)	(nccSTP)	(nccSTP)	(nccSTP)	LB	UB	LB	m	UB	LB	m	UB
230	500	0.1051	5.6969	0.0202	0.0900	0.1253	-22.68	-22.52	-22.34	652	678	708
235	450	0.1369	5.8251	0.0279	0.1167	0.1648	-22.29	-22.13	-21.94	675	702	734
240	400	0.1781	5.9762	0.0390	0.1502	0.2171	-21.89	-21.72	-21.52	705	735	771
245	350	0.2305	6.1807	0.0556	0.1915	0.2861	-21.48	-21.29	-21.07	694	732	772
250	300	0.2777	6.4385	0.0758	0.2221	0.3535	-21.13	-20.91	-20.66	703	747	793
250	300	0.2654	6.7374	0.0725	0.1896	0.3379	-21.25	-20.91	-20.66	654	729	774
250	300	0.2489	7.0231	0.0680	0.1764	0.3169	-21.28	-20.93	-20.65	647	721	765
250	10250	6.8042	7.2901	0.0430	6.7362	6.8472	-20.70	-20.69	-20.68	801	802	803
240	785	0.1853	14.1156	0.0187	0.1423	0.2040	-21.69	-21.42	-21.32	770	819	838
Total			145.8145								998	

The diffusivities determined from the step heat 3He and 4He yields for HT-Win-06-01b are always significantly lower than the diffusivities determined from step heat experiments on the same sample in vacuum, and follow evolutions that are distinct from those determined from the vacuum experiments. The Arrhenius plot of the 3He diffusivities initially follows a declining concave-up trend until ~3.5% cumulative 3He yield, then slowly decreases in sub-parallel arrays during prograde and retrograde heating cycles from 3.5-7% cumulative 3He yields (Fig. 1.6, Fig. 1.7). The Arrhenius plot of the 4He diffusivities differs in shape: it is concave-down to about 5% cumulative 3He yield, after which it follows a roughly linear array about one log unit below the 3He array (Fig. 1.6). The 3He diffusivities settled into a linear array which was suitable for linear regression after ~7% of the cumulative 3He yield was lost (Fig. 1.8b). All remaining diffusivity measurements of the experiment were included in the linear regression, so that diffusivities from 3He loss fractions of about 7-15% were included. The linear regression yields an R^2 of 0.989, an activation energy of 157.1± 5.3kJ/mol, and a closure temperature of 98.1± 8.6°C assuming a cooling rate of 10°C/Ma and considering only the uncertainty in the regression intercept (D_0/a^2).

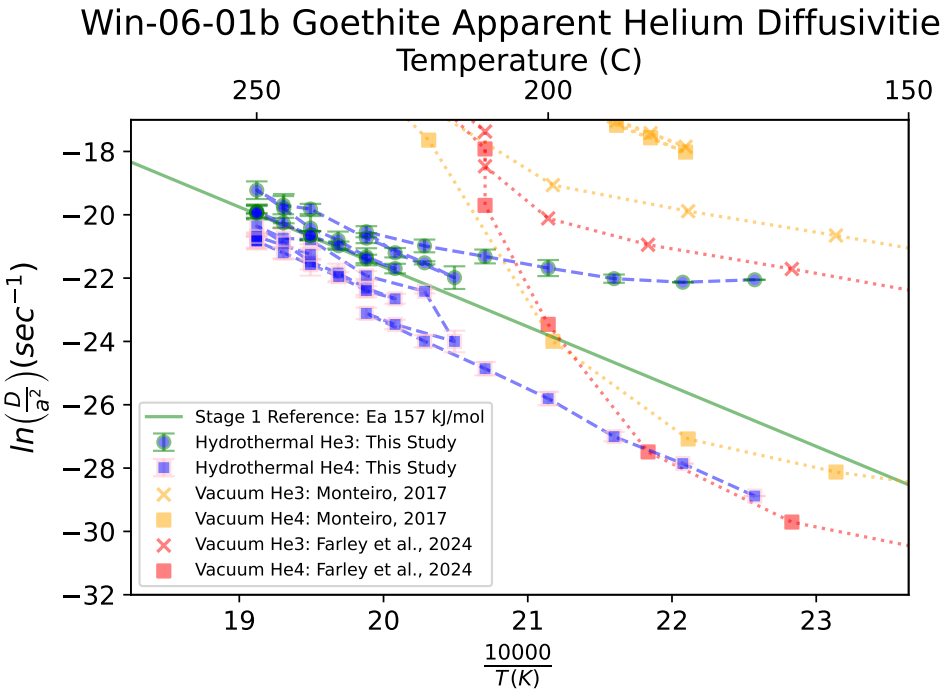


Figure 1.6: Linear regression of the last 12 measurements of 3He diffusivities in HT-Win-06-01b defining the "Stage 1" reference line. Apparent diffusivities under hydrothermal conditions are approximately seven natural log units lower than in vacuum. A diffusion domain corresponding to the Stage 1 reference line has a closure temperature of 98.1±8.6°C assuming a cooling rate of 10°C/Ma. Apparent diffusivities calculated from 4He step heats are sometimes within error of those calculated for 3He; however, they are occasionally resolved at the 2σ level and are about a single log unit lower than apparent diffusivities calculated for 3He after a cumulative fractional yield of 14% 3He.

A delta plot was generated using the intercept and slope parameters from the 3He regression as a reference line, similarly to previous work (Farley, 2018). The 3He apparent diffusivity delta values monotonically decrease until a cumulative 3He loss of 7%, after which the delta values are

within error of Delta = 0: the reference line (Fig. 1.7). A series of low-yield measurements on the first retrograde cycle produce high-error, scattered diffusivities which are omitted for clarity.

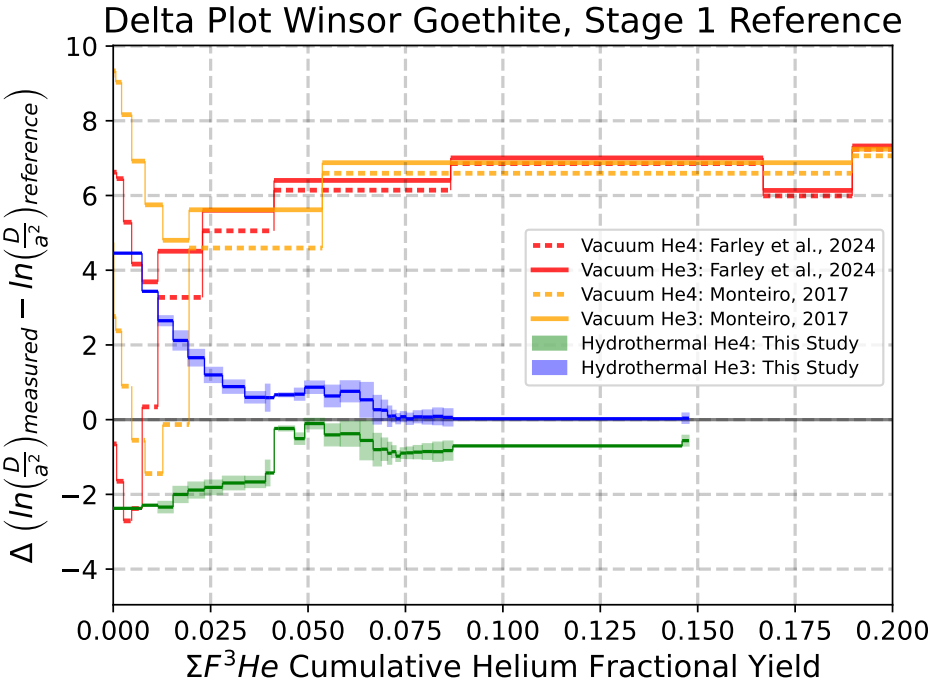


Figure 1.7: Delta plot similar to those presented in Farley (2018), or r/r0 plots in Lovera et al. (1991) and Farley and Flowers (2012). Delta plots are the difference between apparent log diffusivities and a reference line; they are designed to remove the temperature dependence of the dominant process of helium migration to reveal how apparent diffusivities evolve as an experiment proceeds. Six measurements from the water-saturated experiment are omitted for clarity due to low signals and high errors. For visual continuity, the gap left behind is bridged and no uncertainty is shown. 3He monotonically decreases until settling on the reference line from 7–15% 3He fractional yield. 4He diffusivities and delta values are generally highly uncertain but initially show an increase in delta values that settles to a slow increase or plateau about one log unit below the 3He reference line by a 3He fractional yield of 5%, which is consistent with diffusive loss of radiogenic 4He in variably retentive domains. Apparent diffusivities for helium in Win-06-01b in hydrothermal conditions are up to 7 log units lower than the same sample in vacuum.

1.4.2 Kinetic Reanalysis of Vacuum Step Heats

All steps are plotted in Fig. 1.8, for both the kinetic and diffusion reductions, but the fits use only KAF steps 5–22 (185–280 °C) and HDM steps 8–26 (180–296 °C).

An unweighted least-squares fit of $\ln k_i$ against $1/T_i$ for the 35 steps in the fitting window gives

$$E_a = 137.0 \pm 2.7 \text{ kJ mol}^{-1}, \qquad \ln A = 21.91 \pm 0.66 \tag{1.8}$$

KAF alone gives 139.3 ± 6.2 and HDM alone 135.4 ± 2.5 kJ mol^{-1}), in agreement with the cumulative fit. The result does not depend on the fitting window: fitting all steps instead gives $E_a = 122.7$ kJ mol^{-1} and the same ranking of models (§**??**).

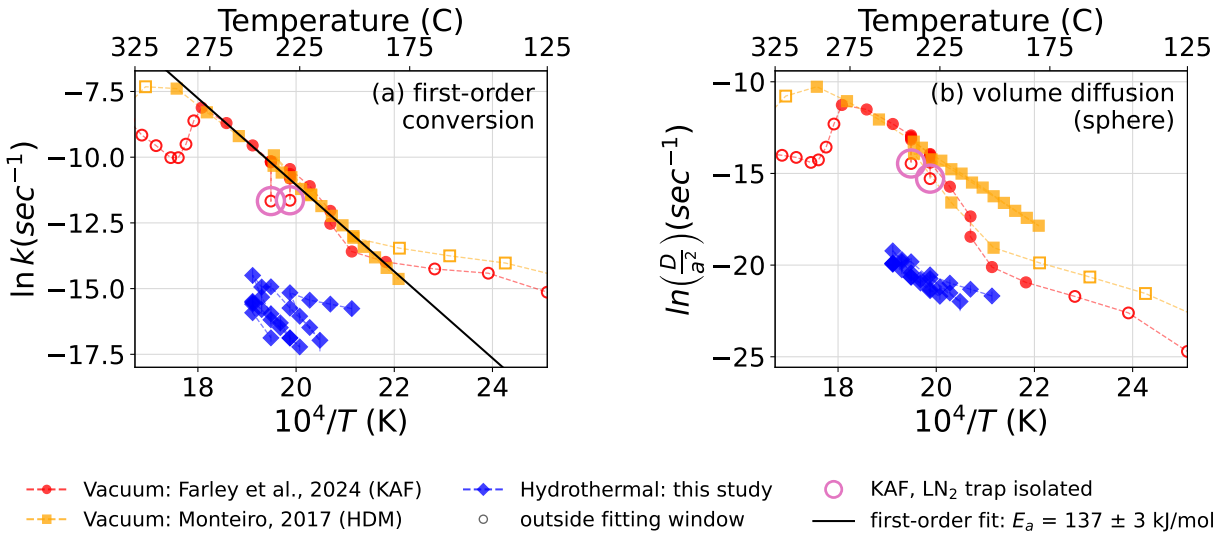


Figure 1.8: The ^{3}He release of every step of the two vacuum step heats of Win-06-01b (red circles: KAF, Farley, H. B. Monteiro, et al. 2024; orange squares: HDM, H. Monteiro 2017) and of the hydrothermal step heat of this study (blue diamonds), reduced in two ways: (a) as first-order conversion rate constants k_i (Eq. (1.3), $\phi = 0.101$), and (b) as apparent diffusivities D/a^2 for a sphere (Fechtig and Kalbitzer, 1966). Only the filled points are used for the linear regressions. Large pink circles mark the two KAF steps with the LN_2 trap isolated. Six low-signal steps of the water-saturated experiment are excluded for clarity. The black line is a first-order Arrhenius fit to filled points of both vacuum experiments. These fitted parameters are given in Eq. (1.8). Below ~180 °C both vacuum experiments depart from the first-order line. Steps with $F > 1 - \phi$ have no defined k and are absent from (a).

The kinetic rate parameters fitted to KAF alone predict the independent HDM schedule with an rms misfit in F of 0.018 over the fitting window. A single-domain spherical diffusion model fits both vacuum step heats worse, even when given the same hematite-inherited fraction ($\phi = 0.101$) as the first-order fit (Table **??**).

KAF steps 1–4 and HDM steps 2–7 (below ∼180 °C) release ^{3}He 0.5–3 orders of magnitude faster than the extrapolated conversion rate (Fig. 1.8). This component amounts to less than 2% of the ^{3}He.

In two KAF steps the trap was isolated and evolved water accumulated in the line. At 230 °C the step rate constant fell to 0.33 times the mean of its neighbors at the same temperature, and at 240 °C to 0.22 times. The rate recovered fully when the trap was restored. Water and ^{3}He were suppressed together. We return to these two steps in §1.5.4, where they calibrate the nucleation model.

1.4.3 Isothermal Discrimination Test at 250 °C

Sequences of isothermal steps can be particularly powerful in discriminating first and second order processes. Diffusion slows quickly as the diffusion front moves inward, depleting the grain rims of helium, whereas the rate of first-order conversion slows less for similar yields. We harnessed the principle to conduct a sensitive test by performing a series of 4 consecutive, long isothermal steps at 250 °C at the end of the water-saturated experiment: three consecutive 5-hour steps and one 170.8-hour step. Using the first three 5 h steps at 250 °C, we invert for both kinetic rate constants and apparent diffusivities, then use each model to project the yield for the 170.8 h step.

Both the first-order and diffusion models yield reproducible results for the three 5-hour calibration steps (Fig. 1.9), but their projections sharply diverge for the long step. Diffusion projects a final fractional cumulative yield of $F = 0.149$ against an observed $F = 0.146$, a much stronger agreement than the first-order model prediction of $F = 0.173$, which overpredicts the helium yield for that step by 46% (Fig. 1.9). The data's uncertainty is too small to affect this result. In the interest of thoroughness, we also tested the contracting-geometry and Avrami models of §**??**, which fit the vacuum data less well. Calibrated the same way, they project cumulative fractional yields within 0.174–0.212, which diverge even more strongly from the data.

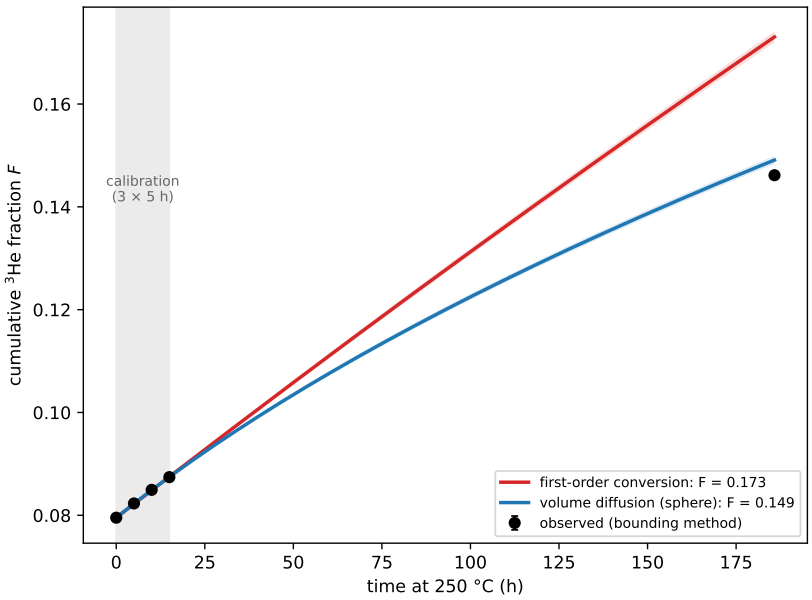


Figure 1.9: The isothermal 250 °C sequence of the water-saturated experiment. Black markers are measured cumulative ^{3}He fraction. The first-order conversion and spherical volume diffusion models are each calibrated on the three 5 h steps (grey band) and projected through the 170.8 h step.

1.5 Discussion

1.5.1 Dehydroxylation Kinetics in Vacuum Step Heats of Win-06-01b

Farley, H. B. Monteiro, et al. (2024) showed that evolved helium in vacuum step heats of Win-06-01b (and many other goethite samples) is controlled by the conversion of goethite to hematite. Here, we present a reanalysis of the vacuum step heat data of Win-06-01b from KAF and HDM to show that the helium release of Win-06-01b is better described by a model of first-order reaction kinetics for goethite dehydroxylation than by a second order diffusion model.

We postulate that the early ^{3}He released faster than the extrapolated conversion rate (§1.4.2) is diffusively lost from the least retentive domains; it corresponds to "Stage 1" helium release of Farley, H. B. Monteiro, et al. (2024). The kinetic reduction yields simpler, more interpretable Arrhenius plots for the vacuum experiments than the diffusion inversion does. The kinetic inversion collapses the prograde steps at the beginning of the vacuum experiments onto a single line that the diffusion reduction instead shows as an irreversible increase in diffusivity (Fig. 1.8). This initial helium release is most easily read as a diffusion-related loss from a broad distribution of small, poorly retentive domains in the goethite. The Arrhenius plot of the kinetic reduction makes it clear that helium release from the initial steps is in excess of the rate of dehydroxylation, which is not obvious from the Arrhenius plot produced with the diffusion reduction. This shift in the dominance from diffusive loss to dehydroxylation-related loss offers a clear explanation for the change in behavior between "Stage 1" and "Stage 2" in vacuum step heat experiments described by Farley, H. B. Monteiro, et al. (2024).

Thermogravimetric studies of goethite dehydroxylation report somewhat higher apparent activation energies, 154 ± 15 to 169 ± 8 kJ mol^{-1}, for a reaction controlled by a two-dimensional phase boundary, with a different mechanism at early stages and low temperature (Goss, 1987). This may simply be due to a difference in samples: goethites from different sources have different particle sizes, morphologies, and aluminum contents, all of which affect the kinetics of dehydroxylation (Langmuir, 1971; Yapp, 1983; Tardy and Nahon, 1985; Majzlan and Navrotsky, 2003).

1.5.2 Helium Release under Water-Saturated Conditions Is Diffusive

The goal of this work was to attempt to slow the goethite dehydroxylation process below the rate of thermal helium loss in unchanging goethite. The thermal schedule included several prograde-retrograde heating cycles which together would have completely dehydroxylated goethite in vacuum, using the vacuum kinetics of Eq. (1.8). Even so, only 14.7% of the ^{3}He was released in the water-saturated experiment; we argue below that the helium loss in the water-saturated experiment is controlled by diffusion rather than by conversion.

Fig. 1.8 compares the kinetic and diffusion inversions of the step heat data. These plots show the opposite trend for the vacuum and hydrothermal experiments: the hydrothermal experiment collapses into a simpler, coherent linear array with the diffusivity reduction, whereas the kinetic inversion of the same data records a continuously declining k for any given temperature as the experiment proceeds, forming a zig-zag pattern. This is the expected behavior for a second-order (diffusion-controlled) process for which helium loss slows more strongly than a first-order process as the experiment proceeds. Fig. 1.8 shows that helium loss is first-order for vacuum experiments and second order for the water-saturated experiment. Section 1.5.4 shows that these observations require the conversion rate at water saturation to be at least $10^{2.5}$ times slower than in vacuum, and that a nucleation control on the dehydroxylation kinetics explains suppression far greater than that.

While it is difficult to prove that decomposition or recrystallization is not playing a role in the results presented in this work, the results presented here suggest that a second-order process (likely volume diffusion) is the dominant mechanism of helium loss for Win-06-01b goethite in water-saturated conditions.

1.5.2.1 Consistency with MDD Systematics

All existing helium step heat data on goethites are from naturally occurring polycrystalline aggregates with unclear and potentially widely variable grain sizes (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024), which likely could follow MDD systematics (Farley, 2018). In MDD systems, apparent bulk diffusivities of a homogeneously distributed diffusant at a single temperature will decrease as an experiment proceeds, or for a system where the activation energy of diffusion is known, the temperature dependence can be removed by referring to the data in relation to a temperature-dependent reference line, as in delta or r/r0 plots (where $\Delta = ln\left(\frac{D}{a^2}\right)_{measured} - ln\left(\frac{D}{a^2}\right)_{reference}$) (Lovera et al., 1991; Farley, 2018). For the activation energy of 157.1 kJ/mol retrieved by regression in this work, the homogeneously distributed 3He step heat data follow the expected MDD systematics and monotonically decrease within a delta plot (Fig. 1.7), which is consistent with volume diffusion in an MDD system.

The strong plateau in the delta plot of the water-saturated goethite experiment and the statistically robust regression over a wide range of fractional cumulative 3He yield (from 7-15% ΣF_{3He} with an R^2 of 0.989) suggest that the frequency factor obtained by the regression may be dominated by a single diffusion domain for $F_c \geq 7\%$ rather than a mix of diffusion domains as is most often observed in the apparent diffusivities for polycrystalline hematite (Farley, 2018).

A multiple diffusion domain inversion of the ^{3}He release (§1.3.2) tests this claim (Table 1.3; Fig. 1.10). A single domain cannot fit the release (Arrhenius $\chi^2_\nu = 27$), and two domains capture the high early diffusivities but leave a poor Arrhenius fit ($\chi^2_\nu = 3.3$). Three domains reduce the yield misfit by more than half ($\chi^2_\nu = 1.08$). With four, the smallest domain splits in two and the fit improves slightly further ($\chi^2_\nu = 1.00$). Every domain added beyond four coincides in size with the dominant domain and leaves the fit unchanged. We therefore adopt four domains as a complete description of the ^{3}He data:

- 0.57% of the ^{3}He at $r/r_0 = 6.8 \times 10^{-4}$;
- 0.66% at $r/r_0 = 2.9 \times 10^{-3}$;
- 1.5% at $r/r_0 = 0.031$;
- 97.2% at $r/r_0 = 1.12$.

The three small domains, holding 2.8% of the ^{3}He in total, are fully degassed during the experiment and account for the high early diffusivities and the gradual approach of the delta plot to the Stage 1 line (Fig. 1.11). The dominant domain releases 12% of its ^{3}He during the experiment and diffuses at 0.80 times the Stage 1 rate. The Stage 1 line is therefore a slightly fast description of the dominant domain, because the bulk-inverted diffusivities on which it was regressed still carry a small contribution from the fast domains. The four-domain model reproduces the Arrhenius array, including its retrograde segments, and the delta plot, although neither is fitted directly. Fits for $N = 1$–5 are shown in the SI (Fig. **??**).

Table 1.3: Multiple diffusion domain fits to the ^{3}He release of HT-Win-06-01b (N domains, $k = 2N - 1$ parameters). RMS is the yield misfit ($\times 10^{-4}$, fraction of total ^{3}He) over the 28 steps above the signal threshold. "Fast" is the ^{3}He fraction in domains with $r/r_0 < 0.1$; f_{main} and r_{main}/r_0 are the fraction and size of the largest-fraction domain. For $N \geq 5$, the dominant domain is split among several domains of identical size.

N	k	RMS	Arrh. χ^2_ν	fast (%)	f_{main} (%)	r_{main}/r_0
1	1	85.30	27.45	0.0	100.0	0.80
2	3	20.20	3.30	2.2	97.8	1.05
3	5	8.45	1.08	2.7	97.3	1.12
4	7	7.81	1.00	2.8	97.2	1.12
5	9	7.81	1.11	2.8	95.1	1.12
6	11	7.81	1.24	2.8	85.6	1.12
7	13	7.81	1.40	2.8	85.3	1.12
8	15	7.81	1.62	2.8	76.8	1.12
9	17	7.81	1.91	2.8	76.5	1.12
10	19	7.81	2.34	2.8	74.6	1.12

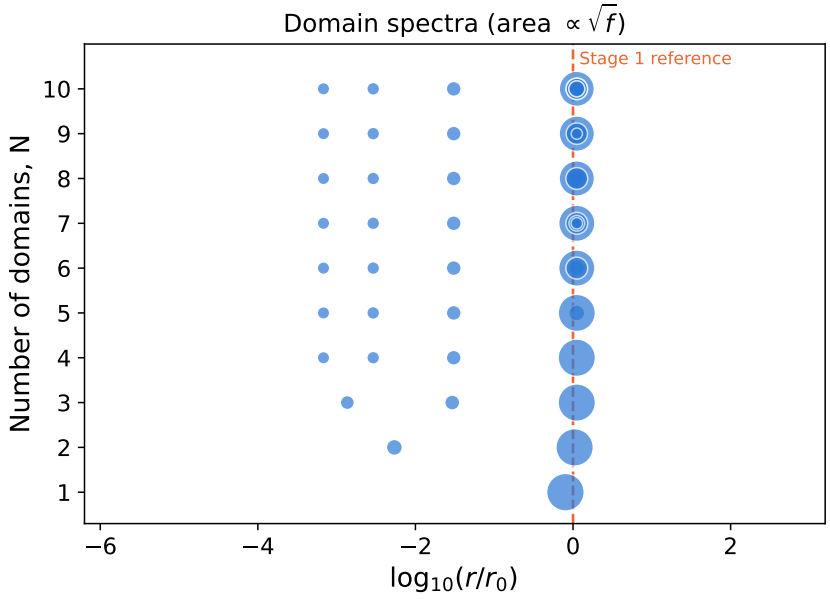


Figure 1.10: Domain spectra of the best-fitting multiple diffusion domain models for $N = 1$–10. Symbol area scales with the square root of the ^{3}He fraction; the dashed line marks the Stage 1 reference domain. The three small domains are unchanged from $N = 4$ onward; domains added beyond $N = 4$ coincide in size with the dominant domain.

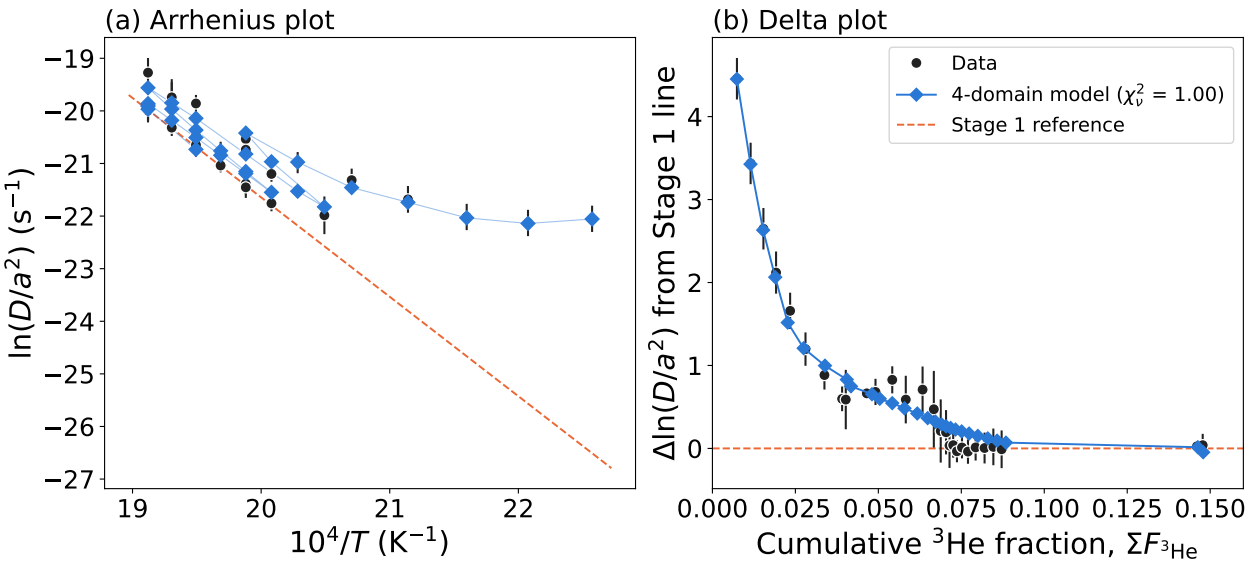


Figure 1.11: (a) Arrhenius plot and (b) delta plot (departure from the Stage 1 line) of the measured ^{3}He diffusivities of HT-Win-06-01b, with the values implied by the four-domain fit to the ^{3}He yields. Neither plot is fitted directly. The six low-signal steps are omitted.

1.5.3 Thermodynamic Model of Goethite Dehydroxylation

The preceding sections establish from helium step heat data alone that the vacuum step heats of Win-06-01b record first order conversion to hematite, whereas the water-saturated step heat records second order diffusion without detectable dehydroxylation. Here we use the thermodynamics of goethite dehydroxylation to explain why water has this effect and to estimate how much solid-state conversion could have occurred in the hydrothermal experiment (§1.5.4).

Goethite breaks down to hematite by the dehydroxylation reaction

$$2\,\alpha\text{-FeOOH} \;\rightleftharpoons\; \alpha\text{-Fe}_2\text{O}_3 + \text{H}_2\text{O}_{(\text{fluid})}, \tag{1.9}$$

whose Gibbs energy change, written per mole of hematite formed, is

$$\Delta G_r(T,P) = G_{\text{hem}}(T,P) + G_{\text{w}}(T,P) - 2\,G_{\text{goe}}(T,P) + \Delta G_{\text{surf}}. \tag{1.10}$$

where ΔG_r is the Gibbs energy change of the reaction and G_{hem}, G_{w}, and G_{goe} are the Gibbs energies of hematite, water, and goethite, respectively. ΔG_{surf} is the surface energy contribution, which must be considered to reproduce observed phase stability (Navrotsky et al., 2008). Goethite is stable where $\Delta G_r > 0$ and hematite plus water where $\Delta G_r < 0$. The values and equations used to evaluate each of these terms are given in the supplement: §**??**.

This model can be used to calculate the stability field for goethite with respect to dehydroxylation to hematite and water in $\log P_{\text{H}_2\text{O}}$–$T$ space (Fig. 1.12). This phase diagram communicates several important features of the goethite–hematite–water system that are relevant to the experiments considered here:

1. Goethite is not stable at the conditions of any of the experiments considered here, in water or in vacuum, in spite of the evidence that it did not react in the water-saturated experiment. Thus, we hypothesize that the reaction is kinetically inhibited for goethite under these conditions.

2. The stability of goethite is strongly dependent on the partial pressure of water below and up to water saturation. Above the saturation pressure, water is liquid and contributes far less to the reaction volume than it does as a gas, becoming a secondary control on goethite's stability.

3. The stability of goethite is strongly dependent on grain size, with smaller grains being more stable than larger grains. At water saturation, bulk goethite is stable below 18 °C, 100 nm goethite below 28 °C, and 10 nm goethite below 108 °C.

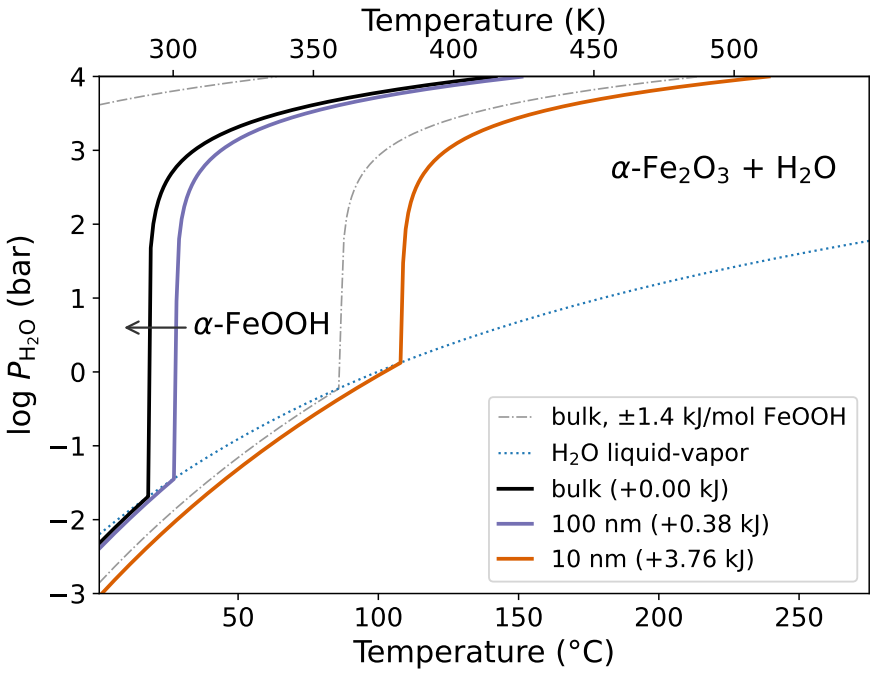


Figure 1.12: Goethite–hematite–water equilibrium in log P_{H_2O}–T space for bulk goethite and for 100 and 10 nm grains with hydrated surfaces, compared at equal surface area per mole of Fe (Eq. (**??**)). Goethite is stable above and to the left of each curve. The dash-dot curves illustrate the absolute uncertainty: it is the bulk stability boundary shifted by the ±1.4 kJ mol^{-1} FeOOH uncertainty of Majzlan, Grevel, et al. (2003). The blue dotted curve shows the water vapor saturation pressure; liquid water is present at and above this curve. The thermodynamic data used to compute the phase diagram are in Table **??**.

Stability alone does not determine what happens in a step heat; the rate of conversion controls whether this equilibrium state is reached or if the reaction proceeds at all, and that depends on the chemical potential of the reaction: $\Delta\mu$. Table 1.4 gives the bulk driving forces at the experimental temperatures, evaluated with IAPWS-95 water. At saturation, $\Delta\mu$ is 6–11 kJ, only 1.6–2.4 times RT. Under the vacuum of a step-heating line ($P_{H_2O} \approx 10^{-8}$ atm) it is 76–106 kJ, about ten times larger. The chemical potential of water is therefore the key variable separating the two experimental settings.

1.5.4 Water Pressure Control on the Hematite-in-Goethite Nucleation Barrier

The hydrothermal data bound how fast goethite could have converted to hematite in water. Under the vacuum kinetics of Eq. (1.8), the hydrothermal schedule gives $\sum_i k_{vac}(T_i)\,\Delta t_i \approx 60$, enough to

Table 1.4: Bulk driving force for hematite formation, $\Delta\mu = -\Delta G_r$ (kJ per mole Fe_2O_3), at water saturation and at $P_{H_2O} = 10^{-8}$ atm.

T (°C)	P_{sat} (bar)	$\Delta\mu_{sat}$	$\Delta\mu_{sat}/RT$	$\Delta\mu_{vac}$	$(\Delta\mu_{vac}/\Delta\mu_{sat})^2$
150	4.76	5.68	1.61	75.8	178
175	8.93	6.88	1.84	83.4	147
200	15.55	8.11	2.06	91.0	126
225	25.50	9.37	2.26	98.5	111
250	39.76	10.66	2.45	106.0	99

convert the sample completely. If water lowers the conversion rate by a factor $R = k_{sat}/k_{vac}$, then even if all of the observed ^{3}He release came from conversion ($\alpha = F/(1-\phi) = 0.147/0.899 = 0.16$), R can be at most $10^{-2.5}$. The release is second order volume diffusion (§1.5.2). Limiting the dehydroxylation fraction to below 1% requires $R \leq 10^{-3.8}$. Here we show that classical nucleation theory, calibrated on the vacuum data alone, predicts an even stronger rate suppression under water-saturated conditions.

Nucleation barrier. In classical nucleation theory (Gibbs, 1878; Volmer and Weber, 1926; Becker and Döring, 1935), reviewed by Christian (2002), Kashchiev (2000) and, for mineral systems, De Yoreo and Vekilov (2003) and Lasaga (1998), a spherical hematite nucleus of radius r has free energy

$$\Delta G(r) = -\frac{4}{3}\pi r^3 \frac{\Delta\mu}{V_{hem}} + 4\pi r^2 \gamma, \tag{1.11}$$

where $\Delta\mu$ is the driving force per mole of Fe_2O_3 (Table 1.4), V_{hem} the molar volume and γ the energy of the interface surrounding the nucleus. Nuclei larger than the critical radius $r^* = 2\gamma V_{hem}/\Delta\mu$ grow, and the energetic barrier to forming one is

$$\Delta G^* = \frac{16\pi\,\gamma^3\,V_{hem}^2}{3\,\Delta\mu^2}. \tag{1.12}$$

In the first-order limit of §1.5.1, the rate constant is a nucleation rate of the form derived by Turnbull and Fisher (1949),

$$k = A\exp\left(-\frac{E_a}{RT}\right)\exp\left(-\frac{\Delta G^*}{k_B T}\right),$$
$$\frac{\Delta G^*}{k_B T} = \frac{C\gamma^3}{T\,\Delta\mu^2}, \qquad C \equiv \frac{16\pi V_{hem}^2}{3k_B} = 1.112\times 10^{15}\ \mathrm{m^6\,K\,J^{-1}\,mol^{-2}}, \tag{1.13}$$

where $A\exp(-E_a/RT)$ describes the atomic mobility and $\exp(-\Delta G^*/k_B T)$ is the equilibrium fraction of sites occupied by critical nuclei. Because the barrier scales as $\gamma^3/\Delta\mu^2$, the tenfold drop in $\Delta\mu$ between vacuum and water saturation raises the energetic barrier ΔG^* a hundredfold (Table 1.4).

Internal nucleation. Goethite dehydroxylates topotactically: hematite nucleates within the parent crystal, inherits its oxygen framework and orientation, and the pseudomorph develops pores and twinned hematite domains (Lima-de-Faria, 1963; Watari, Van Landuyt, et al., 1979; Watari, Delavignette, et al., 1983; Cudennec and Lecerf, 2005). We therefore consider only internal nucleation, in which the new interface of the burgeoning nuclei is between hematite and goethite. We know of no measured or calculated energy for that interface, so γ is replaced by an effective parameter γ_{eff}, calibrated from the data.

Calibration. When Farley, H. B. Monteiro, et al. (2024) isolated the liquid nitrogen trap from the heated goethite, they raised P_{H_2O} in the line and lowered $\Delta\mu$, and the isothermal rate fell by

a factor s (§1.5.1). With A and E_a unchanged, Eq. (1.13) gives

$$\gamma_{\text{eff}} = \left[\frac{T \ln(1/s)}{C \left(\Delta\mu_{\text{off}}^{-2} - \Delta\mu_{\text{on}}^{-2} \right)} \right]^{1/3}, \tag{1.14}$$

with $\Delta\mu_{\text{on}}$ evaluated at $P_{H_2O} = 10^{-8}$ atm and $\Delta\mu_{\text{off}}$ at an upper bound on the water pressure with the trap isolated. This pressure estimate of the water released depends on the volume of the experimental apparatus and the water content emitted by the goethite, which was measured with capacitance manometry (§**??**). This gives $\gamma_{\text{eff}} = 0.16$ J m^{-2} (range 0.14–0.19). Under vacuum the corresponding critical nucleus is smaller than a formula unit ($r^* = 0.08 - -0.11$ nm) and the barrier is only 0.5–1.3 k_BT: conversion in vacuum is effectively barrierless, consistent with the simple Arrhenius behavior of both vacuum experiments. At this scale classical nucleation theory has no literal meaning, so γ_{eff} is best read as the single parameter of a rate law in which $\ln k$ falls as $-1/\Delta\mu^2$, not as an interface energy. At saturation, $r^* \approx 1$ nm and the barrier is 51–129 k_BT.

Translation to water saturation. For the same sample, only $\Delta\mu$ changes between vacuum and saturation:

$$\frac{k_{\text{sat}}}{k_{\text{vac}}} = \exp\left[-\frac{C\gamma_{\text{eff}}^3}{T} \left(\frac{1}{\Delta\mu_{\text{sat}}^2} - \frac{1}{\Delta\mu_{\text{vac}}^2} \right) \right]. \tag{1.15}$$

Applied step by step to the hydrothermal schedule, Eq. (1.15) predicts a total conversion ($\sum_i \frac{k_{\text{sat}}}{k_{\text{vac}}}(T_i)\, k_{\text{vac}}(T_i)\, \Delta t_i$) of $10^{-20.5}$, $10^{-31.4}$ and $10^{-53.7}$ for $\gamma_{\text{eff}} = 0.14$, 0.16 and 0.19 J m^{-2}, and $10^{-67.3}$ to $10^{-15.7}$ (95%) when the uncertainties in the calibration and in the thermodynamic data are propagated (§**??**). The exact values reflect the assumed $\Delta\mu^{-2}$ dependence; what matters is that the model predicts suppression far beyond the $10^{-3.8}$ that the data require.

Assumptions. This is a consistent physical explanation of the data, not a proof. It is not exhaustive. This analysis assumes (i) that the trap-isolated slowdown acts through $\Delta\mu$, rather than, for example, through water blocking reactive sites; (ii) that γ_{eff}, A and E_a are the same in vacuum and in water; (iii) that the barrier depends on the bulk driving force without significant contribution from surface terms (this analysis extended to 10 nm grains raises the predicted conversion only to $\sim 10^{-23}$; §**??**); and (iv) that conversion is solid-state. Conversion by dissolution of goethite and precipitation of hematite from the fluid is not described by Eq. (1.15) (see §1.5.5.1). The calibrated γ_{eff} absorbs contributions from the sample's microstructure and cannot be transferred to other goethites.

1.5.5 Goethite Dehydration and/or Recrystallization During Helium Step Heat Experiments

The aim of helium step heat experiments is to characterize the mechanisms and rates of helium loss in nature to aid in the interpretation of geochronologic and helium geochemical data. The results of this experiment are encouraging in that they differ remarkably from experiments on the same samples in vacuum that are primarily controlled by dehydration processes (Farley, H. B. Monteiro, et al., 2024), but we cannot exclude that this experiment may have been affected to some degree by processes aside from volume diffusion that are unrelated to helium loss in goethite in nature. We consider two processes outside of volume diffusion that could have contributed to the helium released in the water-saturated experiment: (i) recrystallization of goethite or (ii) fluid-mediated conversion of goethite to hematite. If either of these processes affected the helium release, then the apparent diffusivities reported here would represent upper bounds for diffusive helium loss in goethite in nature.

The helium step heat data at water-saturated conditions reported here are better fit with a diffusion model than with a conversion model (§1.5.2). The kinetic analysis of §1.5.1 and the nucleation analysis of §1.5.4 indicate that topotactic dehydroxylation has a nucleation barrier that is insurmountable for experimental conditions and timescales; however, this model has no bearing on fluid-mediated dehydroxylation. Equilibrium thermodynamics predict that goethite is unstable with respect to hematite and water at these conditions (Majzlan, Grevel, et al., 2003) (§1.5.3) and therefore dehydroxylation of goethite in the experiment cannot be outright excluded.

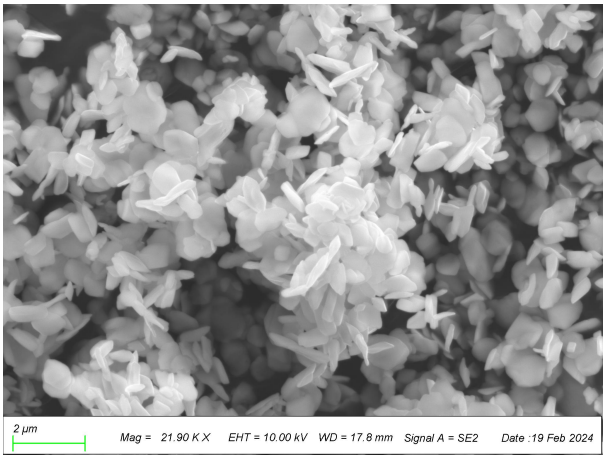
The process of recrystallization is another potential complication in interpreting helium step heats of goethite in hydrothermal conditions. Poorly crystalline synthetic goethite held in hydrothermal conditions is known to recrystallize into more orderly, crystalline goethite aggregates, even at temperatures as low as 125°C in as little as 16 hours (Schwertmann et al., 1985). In this study, natural goethite was held in contact with liquid water and water vapor at hydrothermal conditions as harsh as ~40 bars and 250°C for many days. These conditions appear to have visibly dissolved and reprecipitated silica from the ampoule's inner walls (Fig. 1.4), qualitatively testifying to the ability of the hydrothermal fluid to dissolve and recrystallize rock-forming oxides. It is unquestionable that recrystallization of some goethite could have occurred under these conditions; however, it is unclear if conditions were favorable to recrystallize this specimen of goethite, Win-06-01b, which is amongst the most helium-retentive and crystalline dated goethite samples (Waltenberg, 2012; Farley, H. B. Monteiro, et al., 2024). Less crystalline goethites in nature are commonly recrystallized, which appears to result in helium loss, resetting the U-Th/He age (H. S. Monteiro, P. M. P. Vasconcelos, et al., 2018; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Heim et al., 2006). Recrystallization could be the dominant process controlling helium loss in this experiment, but only if it is faster than diffusive loss, in which case the data presented here would represent new upper bounds for apparent diffusivities for helium in Win-06-01b goethite. Recrystallization might also be faster than volume diffusion of helium in goethite in nature over geologic timescales even if it did not occur in this experiment. In that case, recrystallization could be the primary driver of helium loss in nature regardless of the results of this experiment.

1.5.5.1 Fluid Mediated Conversion of Synthetic Goethite Pigment

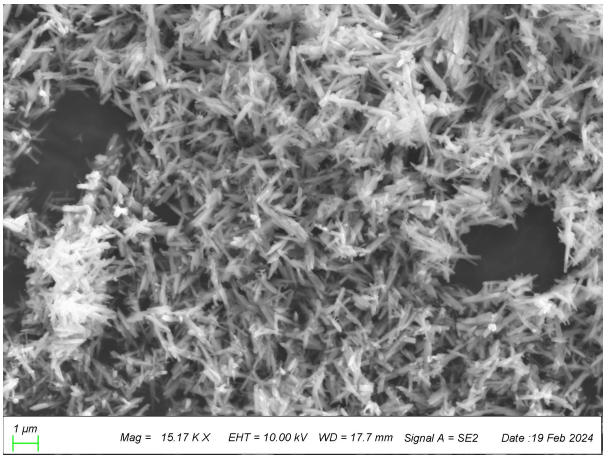
We observed the spontaneous conversion of synthetic goethite pigment (§1.3.3) to hematite when heated in ampoules with excess water at experimental temperatures. Roughly equal proportions of ampoules were completely reacted (red), completely unreacted (yellow), or contained both reacted and unreacted material in isolated patches. In the mixed ampoules, the red and yellow portions were entirely isolated from one another, separated by gas bubbles. Unreacted pigment retains its original acicular habit (Fig. 1.13b). The reacted hematite product is instead composed of equant to platy crystals ~0.5–1 µm across (Fig. 1.13a). The change of shape rules out pseudomorphic dehydroxylation like that observed to occur in goethite heated in vacuum (Figs. 1.2, 1.1, and 1.3), indicating dissolution of goethite and nucleation and growth of hematite from the fluid, a pathway outside the solid-state nucleation model developed in §1.5.4.

Conversion either went to completion within a fluid-connected volume or did not occur at all, even where reacted and unreacted pigment shared the same ampoule and thermal history. We posit that the fluid-mediated reaction requires some threshold of dissolved Fe(III) in the fluid to nucleate hematite, but because the reaction consumes and produces dissolved Fe(III) in equal proportions, it is not self-limiting and proceeds quickly to completion once that threshold is reached.

We conclude that water-saturated conditions do not by themselves exclude fluid-mediated conversion of goethite, even where topotactic conversion is suppressed (§1.5.4). Whether that pathway operates likely depends on the solution chemistry. Still, the helium step heat data for Win-06-



(a) Reacted (red) product



(b) Unreacted (yellow) pigment

Figure 1.13: Secondary electron images of synthetic goethite pigment after hydrothermal treatment at 200–250 °C for about a day (§1.3.3). (a) Material from a completely reacted (red) ampoule: the acicular goethite has been replaced by equant to platy hematite crystals ∼0.5–1 µm across. (b) Material from an unreacted (yellow) ampoule, which retains the unchanged acicular habit of the starting pigment. The change in crystal habit requires dissolution of goethite and precipitation of hematite from the fluid, and the all-or-nothing outcome of the experiments indicates that the reaction is fast once it begins.

01b-HT show no evidence for the abrupt and extreme helium loss that would be expected to accompany a fluid-mediated reaction. The step heat data are instead consistent with a smooth, history-dependent release of helium, as expected for diffusive loss (§1.5.2).

Grain size and bulk thermodynamics cannot explain why the pigment blades reacted when the natural polycrystalline goethite did not. At 1 µm, the surface contribution to ΔG_r is only ∼0.04 kJ, and the bulk driving force at saturation is the same 6–11 kJ for both materials (Table 1.4). The most likely difference is the fluid itself. Commercial iron-oxide pigments are commonly produced by precipitation from iron-salt solutions. They can carry residual soluble salts such as iron and alkali sulfates or nitrates, and they are often surface-treated with organometallic compounds (Legodi and Waal, 2007; Almeida Silva et al., 2017; Xu et al., 2021). Such contaminants would change the pH and ionic strength of the water they are heated in, and with them the solubility of Fe(III). The dissolution–precipitation pathway is limited by how much iron the fluid carries, so a modest increase in Fe(III) solubility could make a nucleation-limited reaction viable. The natural Win-06-01b goethite, by contrast, was heated in distilled water, where near-neutral Fe(III) solubility is very low, potentially offering an explanation for the divergent outcomes.

1.5.6 High Helium Retentivity of Win-06-01b-HT

The apparent diffusivities of the water-saturated experiment are approximately seven natural log units lower than those obtained from the same sample in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) (Fig. 1.8b, Fig. 1.6, Fig. 1.7). This high helium retentivity is explored through three lenses: closure temperature of cooling samples (Dodson, 1973), isothermal diffusion (Wolf et al., 1998), and through comparison with step heats of other polycrystalline samples.

If the slope and intercept on the regression of 3He apparent diffusivities from this experiment are taken to represent a real diffusion domain for goethite, then a closure temperature can be calculated

for this domain. Considering uncertainty only in the intercept (D_0/a^2) term and a cooling rate of 10°C/Ma, the closure temperature for that domain would be 98.1± 8.6°C. The multiple diffusion domain inversion (§1.5.2.1) indicates that the dominant domain, which holds 97% of the ^{3}He, diffuses at 0.80 times the rate of the Stage 1 line. Its closure temperature is correspondingly about 1.6 °C higher, well within the quoted uncertainty; the Stage 1 parameters are used throughout this section.

Goethite is formed near Earth's surface and thus U-Th/He geochronology is most typically applied to date surface processes (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Hofmann et al., 2017; Allard et al., 2018). For crystals of goethite grown and collected near the surface, the concept of a closure temperature is poorly applicable, as it necessarily assumes a cooling rate (Dodson, 1973). In such cases, a better measure of retentivity is an analysis of isothermal retentivity over time (Wolf et al., 1998), which is commonly expressed in terms of t'/t, where t' is the helium age and t is the isothermal holding time (most often the 'real age'). For an isothermal holding time of 100 Ma at 40°C, a diffusive domain with the parameters obtained from the regression of 3He apparent diffusivities of Win-06-01b-HT has a t'/t = 0.968, where t'/t = 1 corresponds to perfect retentivity. Fig. 1.14 shows isothermal holding time against helium age for this same domain for 4 different temperatures. At 25°C, this domain is nearly quantitatively retentive of helium even at billion-year time scales. The results presented here qualitatively support that goethite is highly retentive of helium at surface conditions, even at 100-million-year timescales.

This study of a single sample has no bearing on potential complexities related to aluminum content, radiation damage, or crystal defects on bulk diffusion (Bassal et al., 2022), nor does it permit estimation of the diffusivity at infinite temperature (D0); this would require experiments on controlled grain sizes.

1.5.7 Profiling Radiogenic Helium Loss with $\frac{^4He}{^3He}$ Ratios of Win-06-01b-HT

Bulk-normalized ^{4}He/^{3}He ratios in step heats of irradiated samples measure the rounding of concentration profiles of radiogenic helium by natural helium loss, if the loss mechanism in nature and in the laboratory share the same geometry (Shuster, Farley, et al., 2004; Farley, H. B. Monteiro, et al., 2024). Aside from the first experimental step, bulk-normalized ^{4}He/^{3}He ratios for all experiments begin at vanishingly small values and steeply increase at first, with the rate of increase slowing as the experiments proceed (Fig. 1.15). This behavior is consistent with complete then partial loss of radiogenic helium in nature in the least retentive domains and is observed in all step heat experiments on irradiated goethite (Farley, H. B. Monteiro, et al., 2024). These least retentive domains could represent fine goethite grains, or they could represent interstitial helium trapped between crystallite grain boundaries. Goethite aggregates are known to have interstices that are filled by chemisorbed non-stoichiometric water that represents the earliest water to be lost in vacuum (Naono et al., 1987). Non-stoichiometric hydrogen is known to persist in goethite even after heating to 140°C (Yapp and Poths, 1995). It is therefore reasonable to posit that artificial 3He observed in less-retentive domains may be trapped by non-stoichiometric water or hydroxyl groups within interstices of goethite aggregates. Dehydration of non-stoichiometric interstitial water may be associated with early fractions of degassing 3He from goethite samples during step heats, and could potentially represent the LRD (low retention domain) of Shuster, P. M. Vasconcelos, et al. (2005), and the 3He offgassed during "Stage 1" in vacuum step heats (Farley, H. B. Monteiro, et al., 2024). The high variability of the portion of 3He hosted in early-degassing "Stage 1" fractions within vacuum step heats is sensible considering observations that goethite formed at different conditions is known to have highly variable crystallinity, crystallite ordering, surface area,

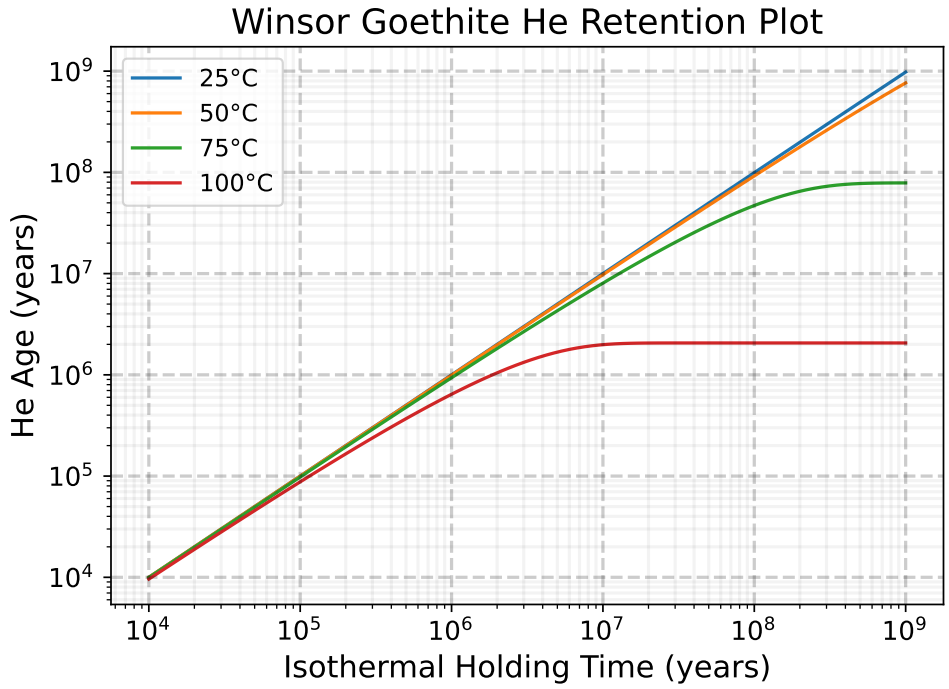


Figure 1.14: Isothermal retention plot (Wolf et al., 1998) for helium in goethite corresponding to the "Stage 1" reference line regression on the hydrothermal experiment. Helium hosted in a diffusion domain with these diffusion parameters would be nearly perfectly retentive at billion-year timescales at 25°C. At 40°C, the equilibrium age for this domain (the helium age at which radiogenic helium production and diffusive loss are equal) is 33.2 billion years, attained after 50.4 billion years.

and non-stoichiometric water contents (Schwertmann et al., 1985).

The hydrothermal experiment records lower bulk-normalized ^{4}He/^{3}He ratios than the vacuum experiments at any given ΣF_{3He} for the duration of the experiment. Similarly, the approach to the ^{4}He/^{3}He plateau varies between the three experiments: a high bulk normalized ^{4}He/^{3}He plateau is most decisively represented in the Farley, H. B. Monteiro, et al. (2024) experiment, whereas the ^{4}He/^{3}He ratios in the step heats of H. Monteiro (2017) transition more gradually before reaching a plateau later in the experiment, while the increase in ^{4}He/^{3}He ratios is slowest for the hydrothermal experiment (Fig. 1.15). This suggests that the geometry of helium loss differs between the hydrothermal and vacuum experiments, and could be explained by the relative dominance of dehydration processes or their mode of spatial propagation as a function of water pressure, where the smoothest ^{4}He/^{3}He spectra are generated by higher water pressures. We interpret this experimental control on the ^{4}He/^{3}He spectra to reflect the relative importance of goethite dehydration and helium diffusion through crystalline goethite in governing the geometry of helium loss from the sample.

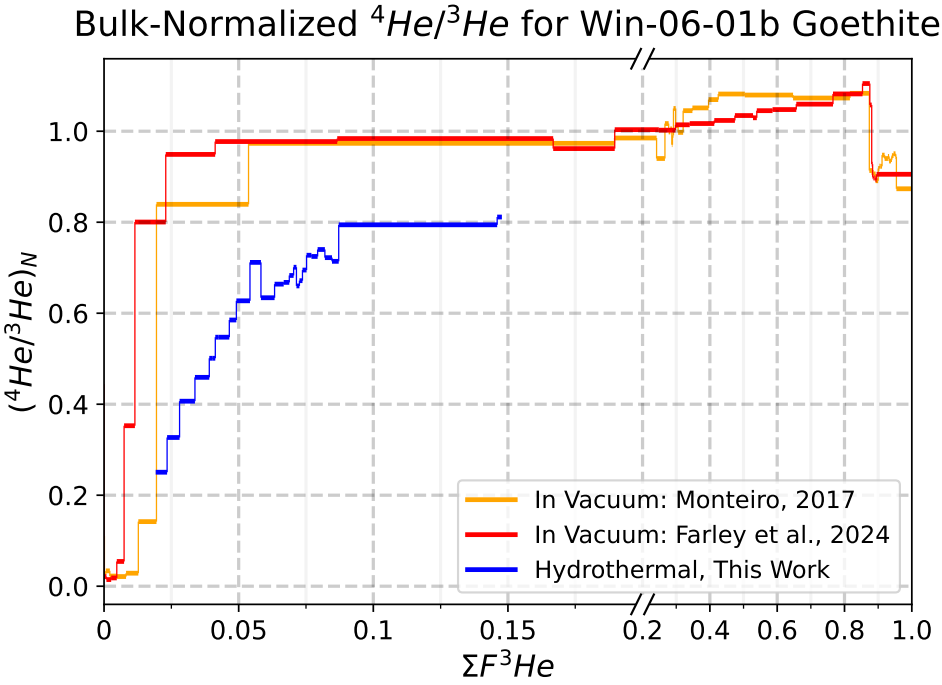


Figure 1.15: Bulk-normalized ^{4}He/^{3}He ratios for Win-06-01b goethite step heats against cumulative fractional ^{3}He yield. The hydrothermal experiment shows lower bulk-normalized ^{4}He/^{3}He ratios at a given ΣF_{3He} for the duration of the experiment, suggesting that ^{4}He/^{3}He spectra are dependent on the partial pressure of water and that the geometry of goethite dehydroxylation differs from the geometry of helium diffusion through goethite. For all experiments, ^{4}He/^{3}He ratios begin at very small values and increase steeply at first, which is consistent with complete then partial loss of radiogenic helium in nature for the least retentive helium domains. Six measurements from the water-saturated experiment are omitted for clarity due to low signals and high errors. For visual continuity, the gap left behind is bridged. The horizontal axis changes scale at $\Sigma F_{3He} = 0.2$ (double slash) to emphasize the portion of the plot where data exist for the water-saturated experiment.

If the geometry of helium loss due to dehydration and due to diffusion differ, then the ^{4}He/^{3}He

spectra of goethite measured in vacuum may not straightforwardly encode the thermal history of the sample in the way that such data faithfully encode this history in phases such as hematite or apatite (Farley and McKeon, 2015; Shuster, Farley, et al., 2004). More ^{4}He/^{3}He spectra of goethite under hydrothermal conditions are needed to investigate this hypothesis further.

These ratios also bear on the natural thermal history of Win-06-01b. Throughout the range of $\Sigma F_{3\mathrm{He}}$ that defines the Stage 1 reference line (7–15%), step ^{4}He/^{3}He ratios remain at 0.67–0.82 of the bulk value (Table 1.2). Radiogenic ^{4}He is depleted relative to the uniformly distributed ^{3}He in this domain, so even this retentive domain lost helium in nature. A domain with the Stage 1 parameters is nearly quantitatively retentive at Earth-surface temperatures (§1.5.6), so this loss is difficult to attribute to thermally activated diffusion at surface conditions. Either Win-06-01b experienced temperatures well above surface conditions (perhaps it grew in a hydrothermal setting or experienced burial heating), or non-thermal processes such as fluid-mediated recrystallization caused partial helium loss in nature. Section 1.5.8 tests whether thermally activated diffusion alone, within the domain structure inferred from the ^{3}He release, can reproduce the measured spectrum.

1.5.8 Thermal History of Win-06-01b

We begin the exercise by limiting our exploration of potential thermal histories to geologically reasonable and simple paths. With the four-domain structure of §1.5.2.1 held fixed, we fitted steady-cooling, burial-and-exhumation, and isothermal thermal histories to the ^{4}He/^{3}He spectrum of HT-Win-06-01b (§1.3.2). Previous studies constrain our exploration of the burial-and-exhumation paths. Gibson and Stüwe (2000) and Mitchell et al. (1998) interpret regional cooling from roughly 50–70 °C after the mid-Cretaceous from their apatite fission-track studies. Mitchell et al. (1998) additionally interpret, from limited 1.5 km thick sediment accumulation in Triassic intramontane basins, that parts of the ranges were near the surface in the early Mesozoic. Table 1.5 gives the best fit of each family, Fig. 1.16 compares the best steady-cooling and burial predictions with the spectrum, and Fig. 1.17 shows their χ^2 maps on a common color scale.

Table 1.5: Thermal histories fitted to the ^{4}He/^{3}He spectrum of HT-Win-06-01b with the four-domain structure fixed. Each row gives the best path with a predicted He age of 130–180 Ma along with the reduced χ^2 value ($\nu = 22$ for steady cooling and the isothermal hold, $\nu = 21$ for burial). For steady cooling, $T_{\max} = T_0$ is reached at crystallization, so $t_{\mathrm{peak}} = t_{\mathrm{form}}$. For the isothermal hold, $T_{\max} = T_{\mathrm{iso}}$ throughout. Burial crystallization ages are fixed within each row. The last three columns give the ranges of the age-consistent paths within the joint 1σ region about the best fit (see Eq. (1.7)).

Path	t_{form} (Ma)	t_{peak} (Ma)	$T_{\max}$ (°C)	χ^2_ν	t_{He} (Ma)	t_{form} (1σ)	t_{peak} (1σ)	$T_{\max}$ (1σ)
Linear cooling	150	150	78	2.00	132	150–175	150–175	76–80
Burial	180	70	61	5.36	164	fixed	60–82	58–65
Burial	195	80	61	5.04	177	fixed	68–98	58–65
Burial	210	115	71	6.53	180	fixed	105–130	70–74
Isothermal hold	150	–	41	18.5	141	140–190	–	39–43

Steady cooling. Linear cooling to 25 °C fits the spectrum best. The best age-consistent path starts at 78 °C at 150 Ma ($\chi^2_\nu = 2.00$), and the age-consistent 1σ range is crystallization at 150–175 Ma at 76–80 °C, followed by cooling at ∼0.3 °C Myr^{-1}. Such paths pass through ∼55 °C at 90 Ma, inside the apatite fission-track window. The spectrum alone prefers somewhat younger,

hotter paths: the unconstrained minimum ($\chi^2_\nu = 1.68$; 89 °C at 80 Ma) predicts a He age of 68 Ma, and the unconstrained 1σ region lies entirely at ages younger than 130 Ma (Fig. 1.17b).

Burial and exhumation. Burial paths fit the spectrum less well ($\chi^2_\nu = 5.0$–6.5), but they agree with the independent regional constraints of Gibson and Stüwe (2000) and Mitchell et al. (1998). For crystallization at 180 and 195 Ma, the best age-consistent paths peak at 61 °C at 70 and 80 Ma, with 1σ ranges of 58–65 °C between 98 and 60 Ma (Fig. 1.17a). These peaks fall in or just after the apatite fission-track window, within the 50–70 °C range of Gibson and Stüwe (2000) and at the <60 °C limit reported by Mitchell et al. (1998). For crystallization at 210 Ma, only hotter and earlier peaks (70–74 °C at 105–130 Ma) give an age within the measured range, and they fit worse ($\chi^2_\nu = 6.5$).

Isothermal hold. The spectrum constrains only the product of diffusivity and time, so the χ^2 minimum is a ridge from ∼47 °C at 50 Ma to ∼35 °C at 500 Ma, along which $\chi^2_\nu \approx 18.5$ throughout. No isothermal path reproduces the shape of the spectrum: they underpredict the ratios of the early steps and overpredict those of the late steps. We do not consider this family further.

Where the families differ. The steady-cooling and burial predictions differ mainly in the earliest steps ($\Sigma F_{^3\mathrm{He}} \lesssim 0.035$; Fig. 1.16), which sample the outer rim of the dominant domain. On the burial path the sample sits at 20 °C for the last 50 Myr, long enough for radiogenic ^{4}He to reaccumulate in the rim, so the early ratios are overpredicted. Steady cooling keeps the rim depleted until geologically recent times, with diffusive loss declining monotonically toward the present. Elsewhere both leave the same residual pattern: ratios underpredicted between $\Sigma F_{^3\mathrm{He}} = 0.04$ and 0.06, and overpredicted in the two longest high-temperature steps.

Consistency with diffusive loss. These two families of thermal paths encode very different geologic origins. We note that any number of arbitrarily complex thermal paths can be evaluated against the data with this workflow, but a thorough search for viable thermal paths is beyond the scope of this study, and is an ill-posed problem for a single sample. The main purpose of this exercise is to demonstrate that the data we present here are consistent with diffusive loss of radiogenic helium in nature. We do not claim to have found the unique thermal history of the sample, but have demonstrated that plausible thermal histories reproduce our results reasonably well.

1.6 Conclusions

Predicting the degree of helium retention and loss to guide interpretation of U-Th/He dates of natural goethite based on factors such as grain size, crystallinity, radiation damage, defect density, impurities, and aluminum substitution remains challenging. Previous efforts to understand helium loss in natural goethite with step heating experiments in vacuum have instead queried the kinetics of dehydration (Farley, H. B. Monteiro, et al., 2024). Because goethite dehydration is suppressed at high water pressure, helium step heats under hydrothermal conditions are more likely to interrogate the processes controlling radiogenic helium loss in natural goethite. The method presented in this study leverages the high helium permeability of silica glass to easily conduct helium step heat experiments under water-saturated conditions, achieving internal water pressures up to 86 bars at 300°C using standard ultra-high vacuum experimental setups (Farley, Reiners, et al., 1999).

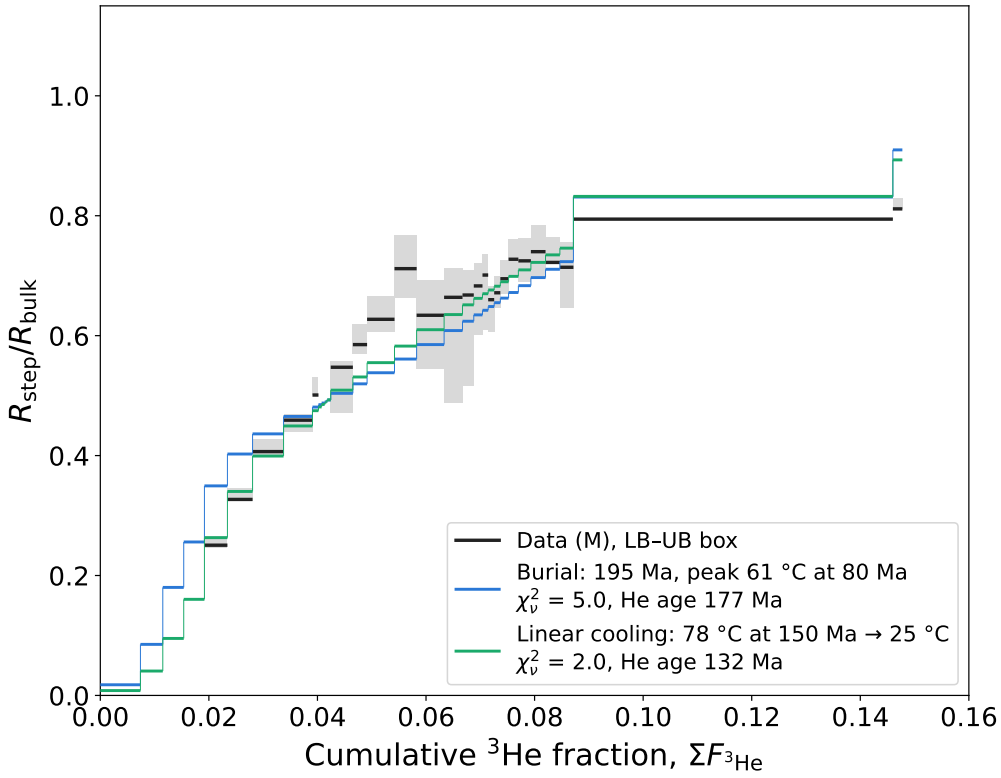


Figure 1.16: Bulk-normalized ^{4}He/^{3}He spectrum of HT-Win-06-01b (black, bounding-method LB–UB boxes) with the best age-consistent burial path for crystallization at 195 Ma (blue) and the best age-consistent steady-cooling path (green), for the four-domain structure.

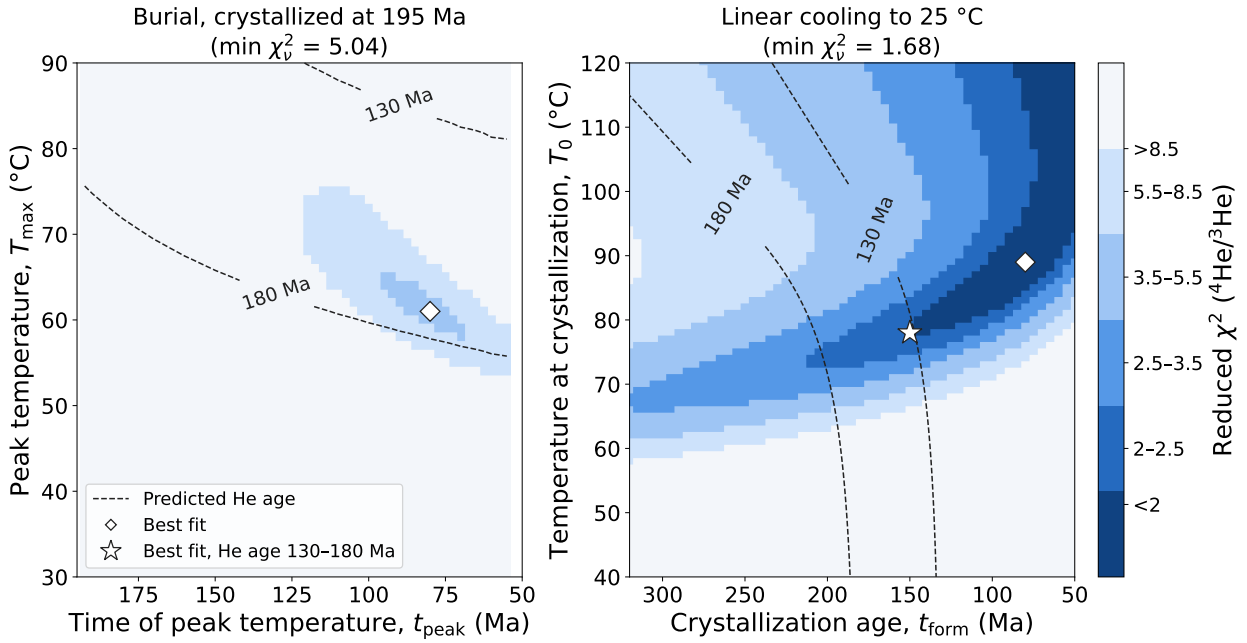


Figure 1.17: Reduced χ^2 of the ^{4}He/^{3}He spectrum for (a) burial paths with crystallization at 195 Ma, as a function of the time and magnitude of peak temperature, and (b) steady cooling to 25 °C, as a function of crystallization age and initial temperature. Both panels share one color scale, anchored to the overall minimum. Dashed lines show the predicted bulk He age at 130 and 180 Ma, diamonds the best fit, and the star the best fit with a He age of 130–180 Ma.

Natural crystalline goethite does not develop the visible dehydration-induced contraction cracks when heated at water-saturated conditions like vacuum-heated goethite does.

The experiment conducted on irradiated Win-06-01b represents the first helium step heat experiment for goethite at water-saturated conditions. Vacuum step heats of the same sample follow first-order conversion to hematite, whereas the hydrothermal release shows the history and time dependence of diffusion. The hydrothermal experiment gives apparent diffusivities approximately seven natural log units below those obtained from the same sample in vacuum. The corrected ^{3}He array is linear between 7–15% cumulative yield, with an activation energy of 157.1± 5.3kJ/mol. Extrapolated to surface conditions, a domain with these parameters is essentially quantitatively retentive of helium at billion-year timescales at 25°C. The ^{4}He data require that this domain experienced helium loss in nature. A four-domain diffusion model fitted to the ^{3}He release shows that 97% of the ^{3}He resides in a single domain and can account for this loss by thermally activated diffusion alone. Within that model, steady cooling from crystallization at 76–80 °C at 150–175 Ma to the present surface temperature fits the ^{4}He/^{3}He spectrum best, favoring crystallization from warm, possibly hydrothermal, fluids. Burial of surface-formed goethite to a Late Cretaceous peak of 58–65 °C, consistent with regional apatite fission-track constraints, remains a viable alternative.

The ^{4}He/^{3}He spectrum observed for this sample is distinct from that of the same sample in vacuum, suggesting that the geometry of helium loss induced by goethite dehydration is distinct from that of diffusion. More hydrothermal step heat experiments of goethite are necessary to confirm this hypothesis.

A thermodynamic and kinetic analysis of goethite dehydroxylation explains why water seems to stabilize goethite up to 250°C at experimental timescales. The hydrothermal release itself requires

solid-state conversion to be at least $10^{2.5}$ times slower in water than in vacuum. At water saturation, the roughly tenfold smaller driving force raises the barrier to hematite nucleation enough to suppress solid-state conversion far below this limit. Fluid-mediated conversion, which transformed synthetic goethite pigment in similar conditions, is rapid and would be accompanied by abrupt and extreme helium loss, which is not observed in the hydrothermal step heat data. This divergent outcome may be explained by different solution chemistry between the two experiments, which would affect the availability of aqueous Fe(III) that supplies the reaction. Recrystallization, however, cannot be excluded. If it contributed, the diffusivities reported here would be upper bounds and the retentivity a lower bound. Distinguishing these possibilities, and separating the effects of crystallite size, chemical composition, and radiation damage, will require further hydrothermal experiments on irradiated single-crystal goethite of controlled size. The method described here makes such experiments straightforward.

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